

Quality control of seismic data and metadata for AdriaArray stations

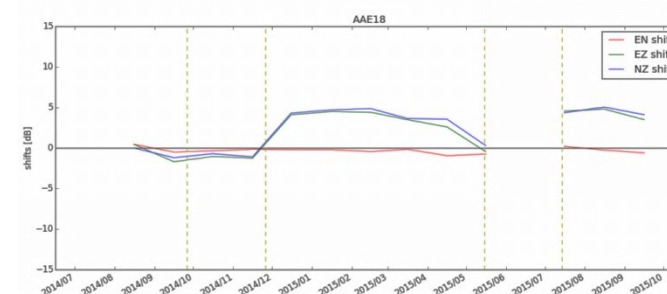
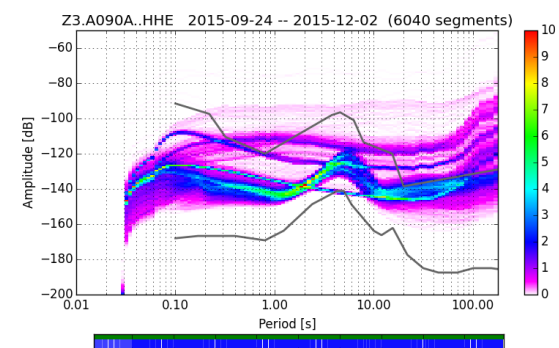
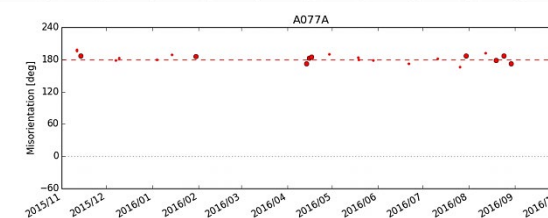
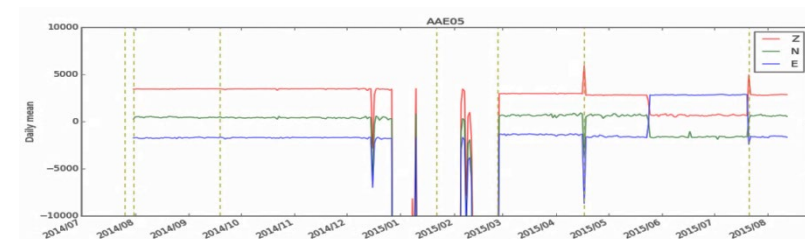
Luděk Vecsey

Quality control and State of Health



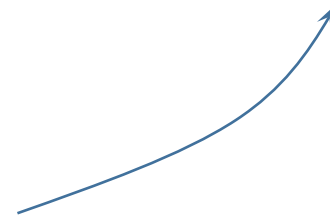
Data availability
Metadata
Signal clipping
Drift of sensor mass position
Amplitude gain imperfection
Noise in signals
Glitches in signal
Sensor orientation
Reversed or interchanged
components
Timing issues

...



Metadata health

<https://github.com/PetrColinSky/AdriaArray>



- list of AdriaArray stations: Petr's tables, _ADARRAY (Mar 29) (broad-band, permanent + temporary, 2022+)
- metadata downloaded from EIDA (Mar 29-30)
- StationXML validation, ObsPy warnings
- compilation of attributes from StationXMLs to CSV table
- corner periods computed from the metadata responses
- checks for P&Z consistency
- cross-checks of corner periods from metadata and independent list

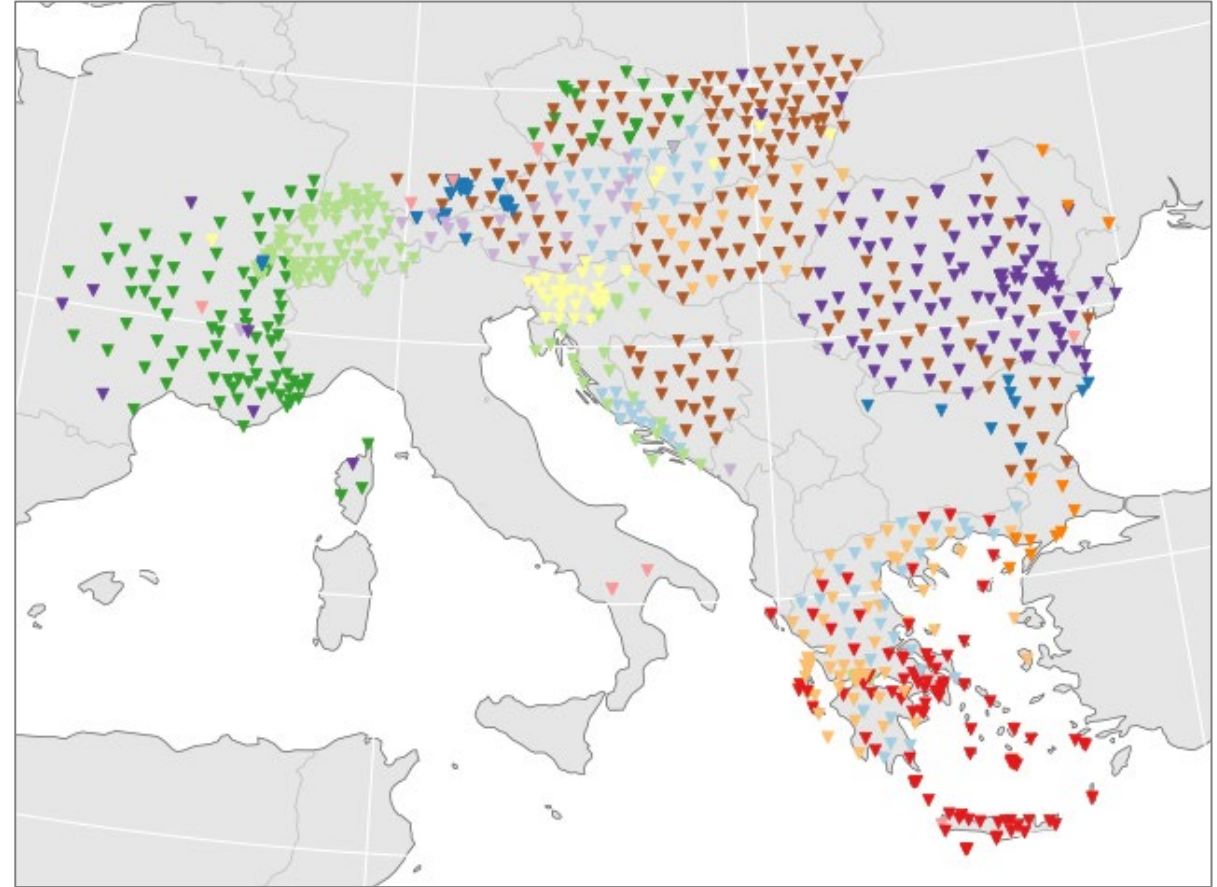
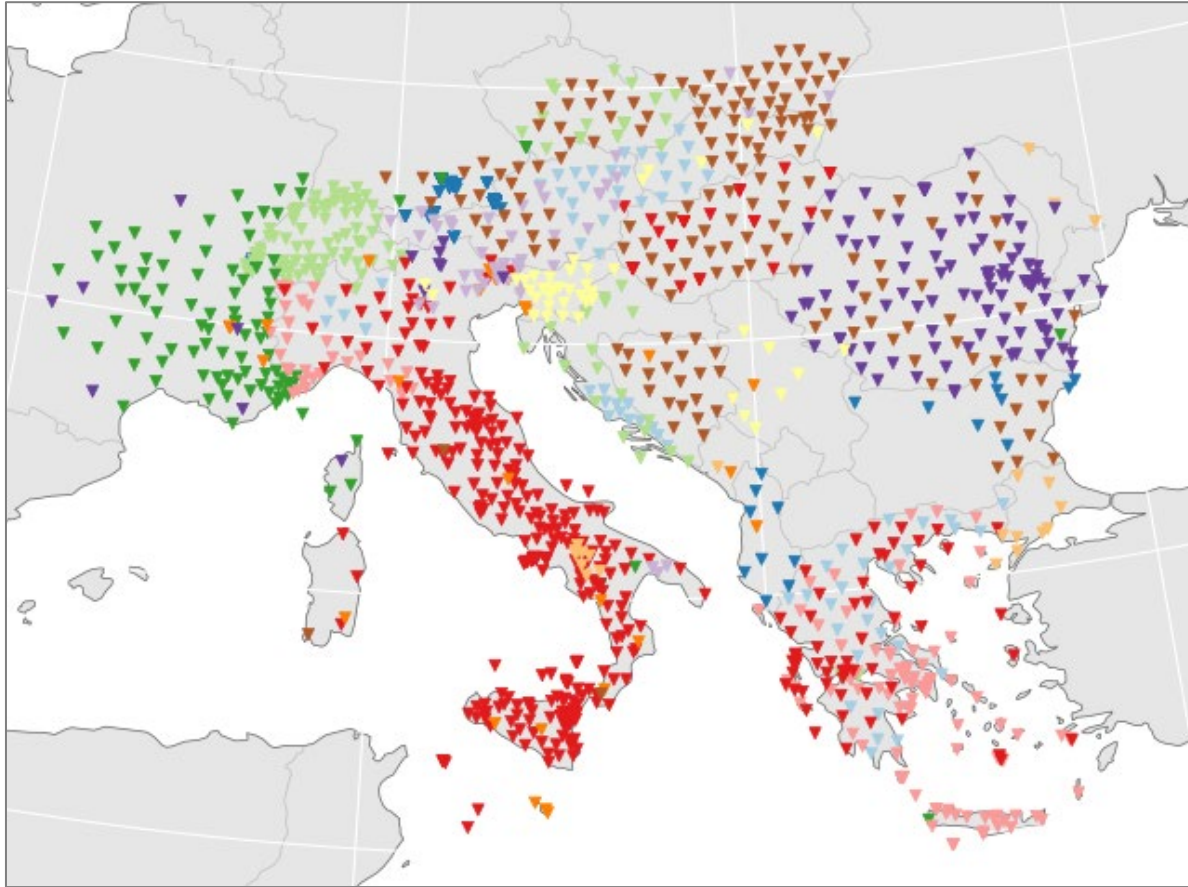
Tested stations

Downloaded metadata from EIDA: 1497 stations
Scripts for checking metadata written in **ObsPy**

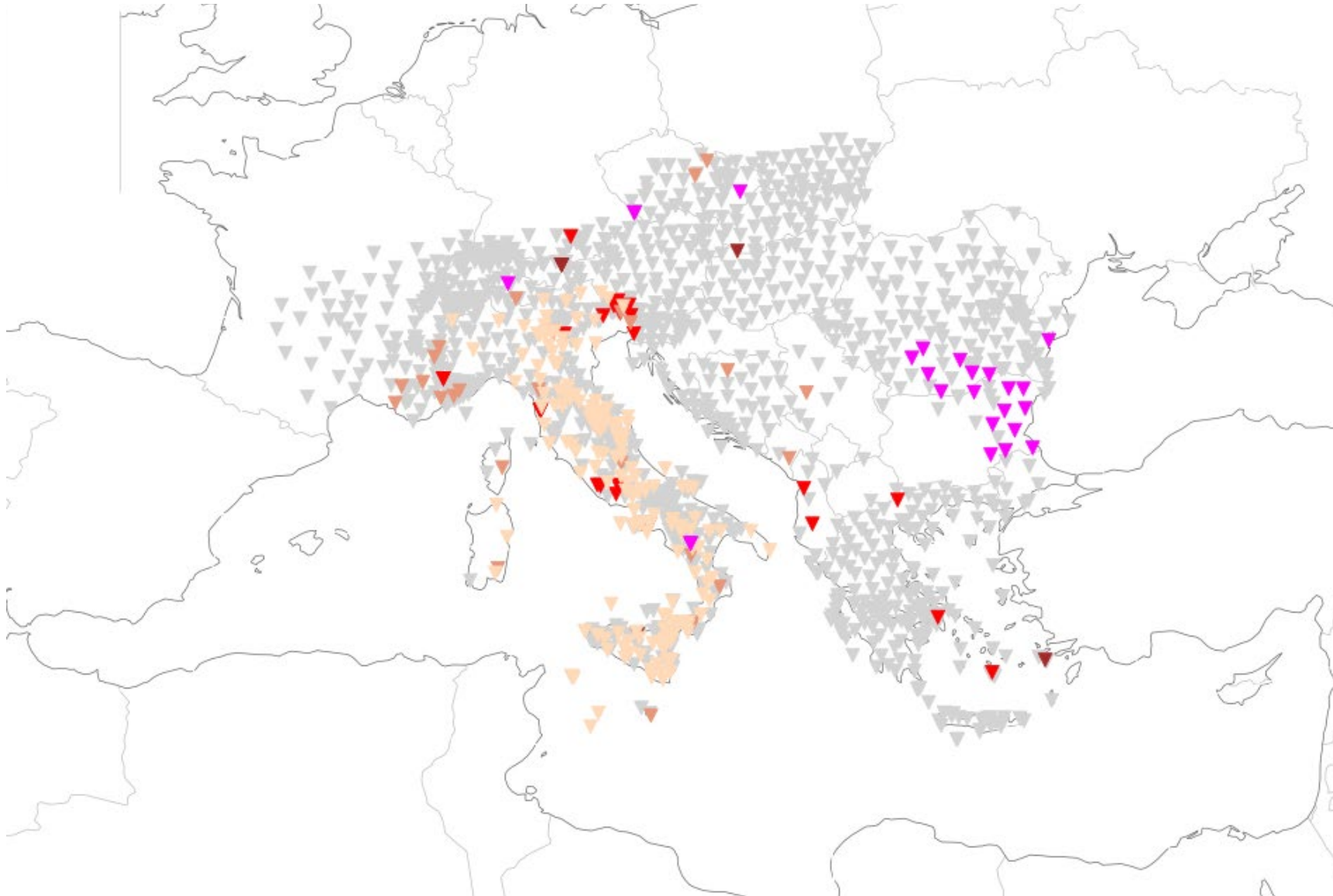
Petr's tables

30.3.2023 00:00

_ADARRAY



Metadata issues



- ▼ no valid FDSN StationXML schema (192/1497)
- ▼ warnings while ObsPy reading metadata (37/1497)
- ▼ no Poles in the sensor response stage (22/1497)
- ▼ Poles issues: complex not conjugated, not negative real parts (5/1497)
- ▼ other errors in metadata (27/1497)

total:
warnings 15.3 %
errors 3.6 %

Wrong band-code naming in metadata

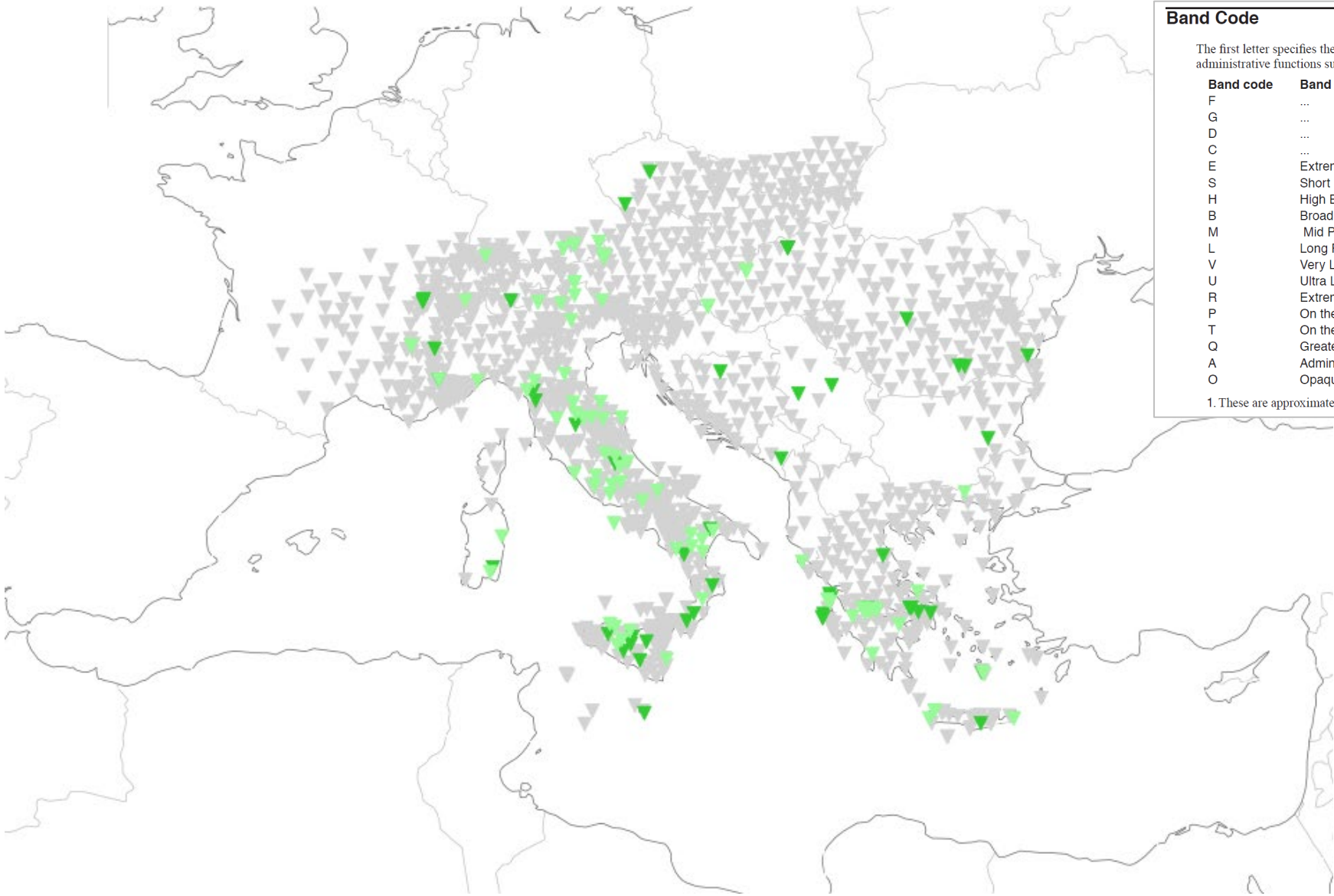
sampling
frequency
20 Hz

+

corner
period
60 s

=

band
code
BHZ



Band Code			
The first letter specifies the general sampling rate and the response band of the instrument. (The "A" code is reserved for administrative functions such as miscellaneous state of health.)			
Band code	Band type	Sample rate (Hz)	Corner period (sec)
F	...	≥ 1000 to < 5000	≥ 10 sec
G	...	≥ 1000 to < 5000	< 10 sec
D	...	≥ 250 to < 1000	< 10 sec
C	...	≥ 250 to < 1000	≥ 10 sec
E	Extremely Short Period	≥ 80 to < 250	< 10 sec
S	Short Period	≥ 10 to < 80	< 10 sec
H	High Broad Band	≥ 80 to < 250	≥ 10 sec
B	Broad Band	≥ 10 to < 80	≥ 10 sec
M	Mid Period	> 1 to < 10	
L	Long Period	≈ 1	
V	Very Long Period	≈ 0.1	
U	Ultra Long Period	≈ 0.01	
R	Extremely Long Period	≥ 0.0001 to < 0.001	
P	On the order of 0.1 to 1 day ¹	≥ 0.00001 to < 0.0001	
T	On the order of 1 to 10 days ¹	≥ 0.000001 to < 0.00001	
Q	Greater than 10 days ¹	< 0.000001	
A	Administrative Instrument Channel	variable	NA
O	Opaque Instrument Channel	variable	NA

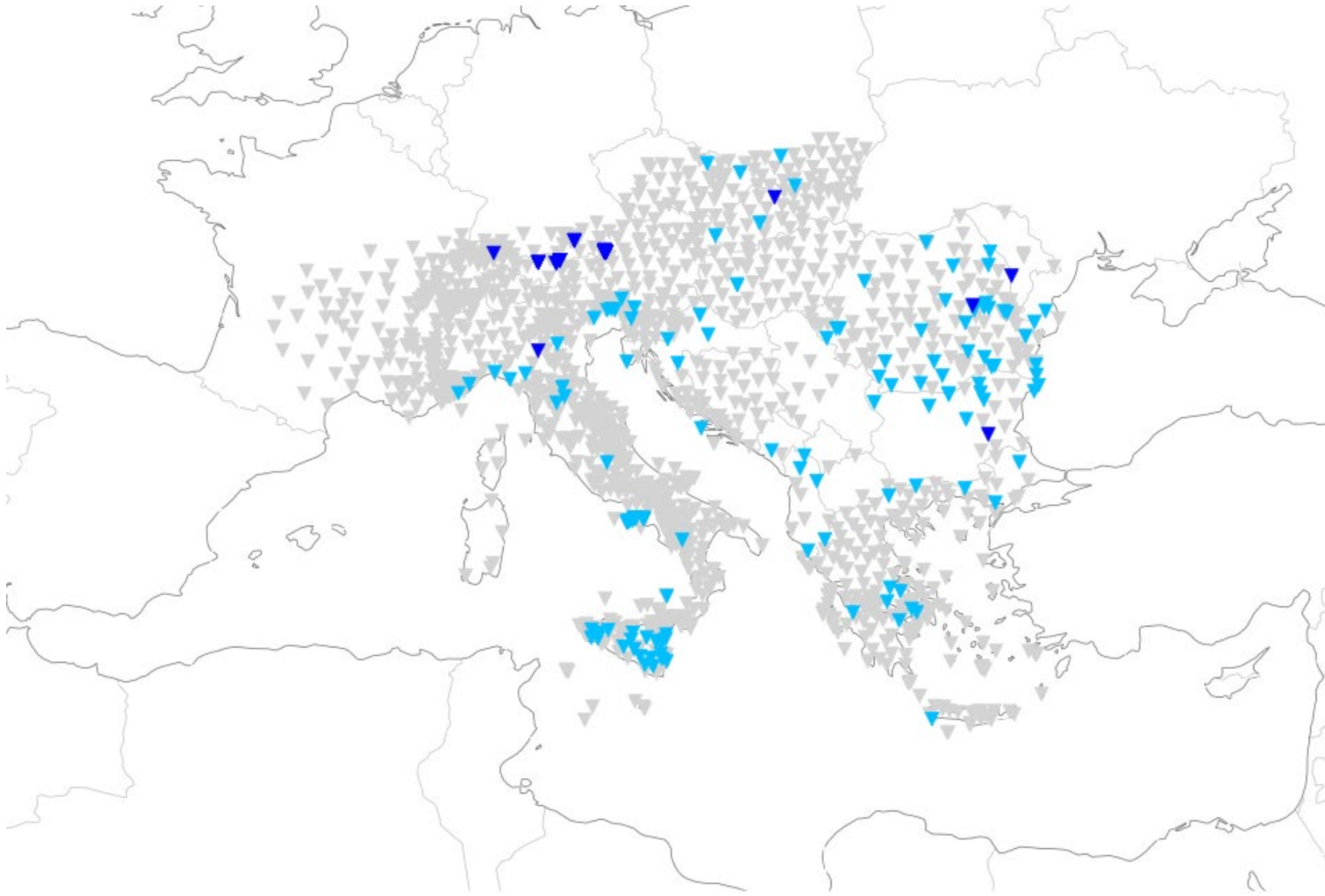
1. These are approximate values. The sample rate should be used for the correct Band Code.

[SEED manual](#)

- ▼ wrong band code in epochs before 2022 (77/1497)
 - ▼ wrong band code during AdA 2022+ (48/1497)
- total: 8.4 %

Cross-checks of corner periods from metadata and independent list

- checks over **recent** epochs in metadata



▼ different corner periods
in metadata and the list
(131/1497)

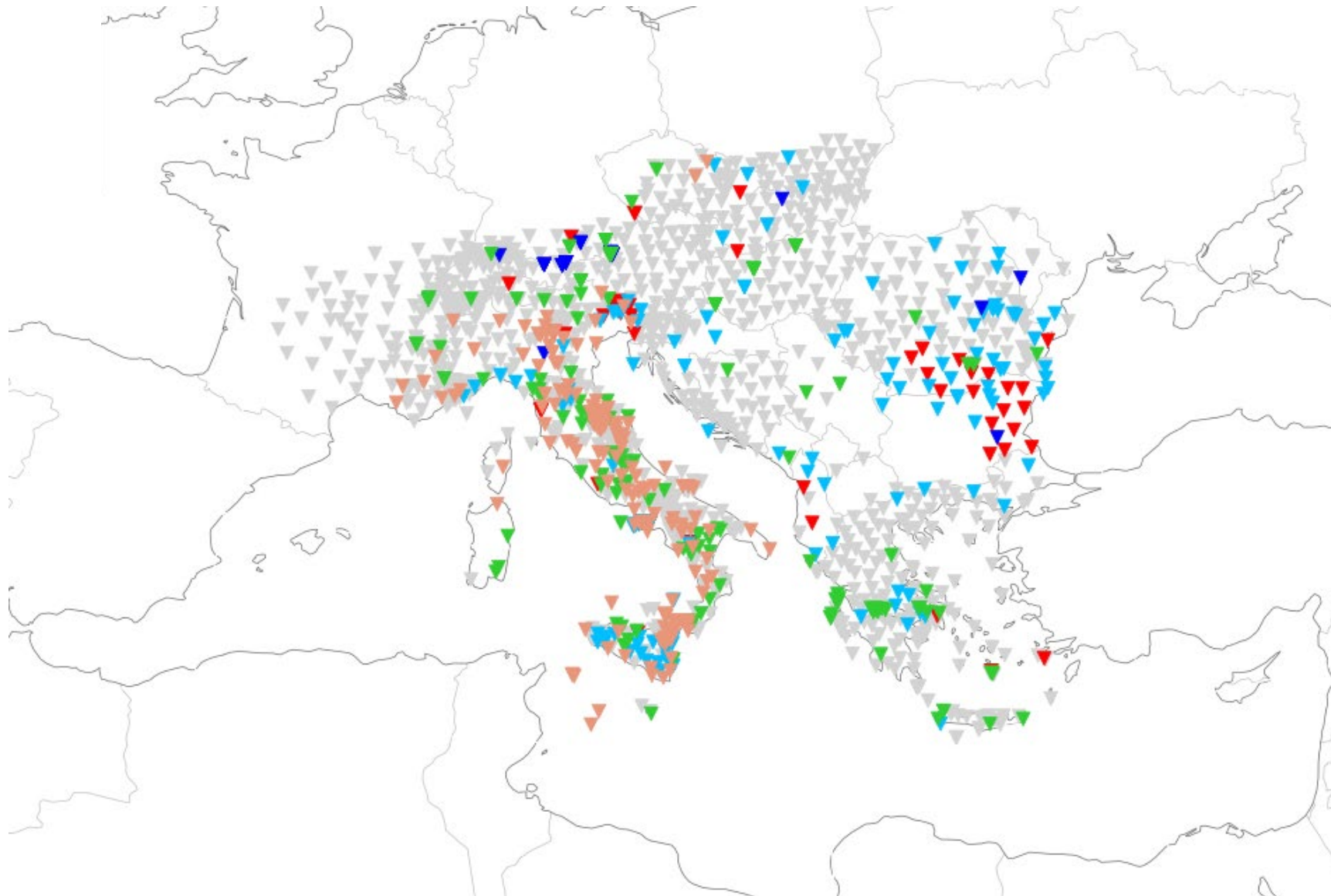
▼ large difference,
**switched broad-band
and short-period**
(15/1497)

total: 9.8 %

Condition to mark corner
periods different:

> 10 % AND > 5 s

All issues in metadata



▼ metadata warnings
(15.3 %)

▼ errors in metadata
(3.6 %)

▼ wrong band code
(8,4 %)

▼ different corner periods
in metadata and the list
(8.8 %)

▼ switched broad-band
and short-period
(1.0 %)

total: 37 %

Daily means – fast and powerful tool for QC

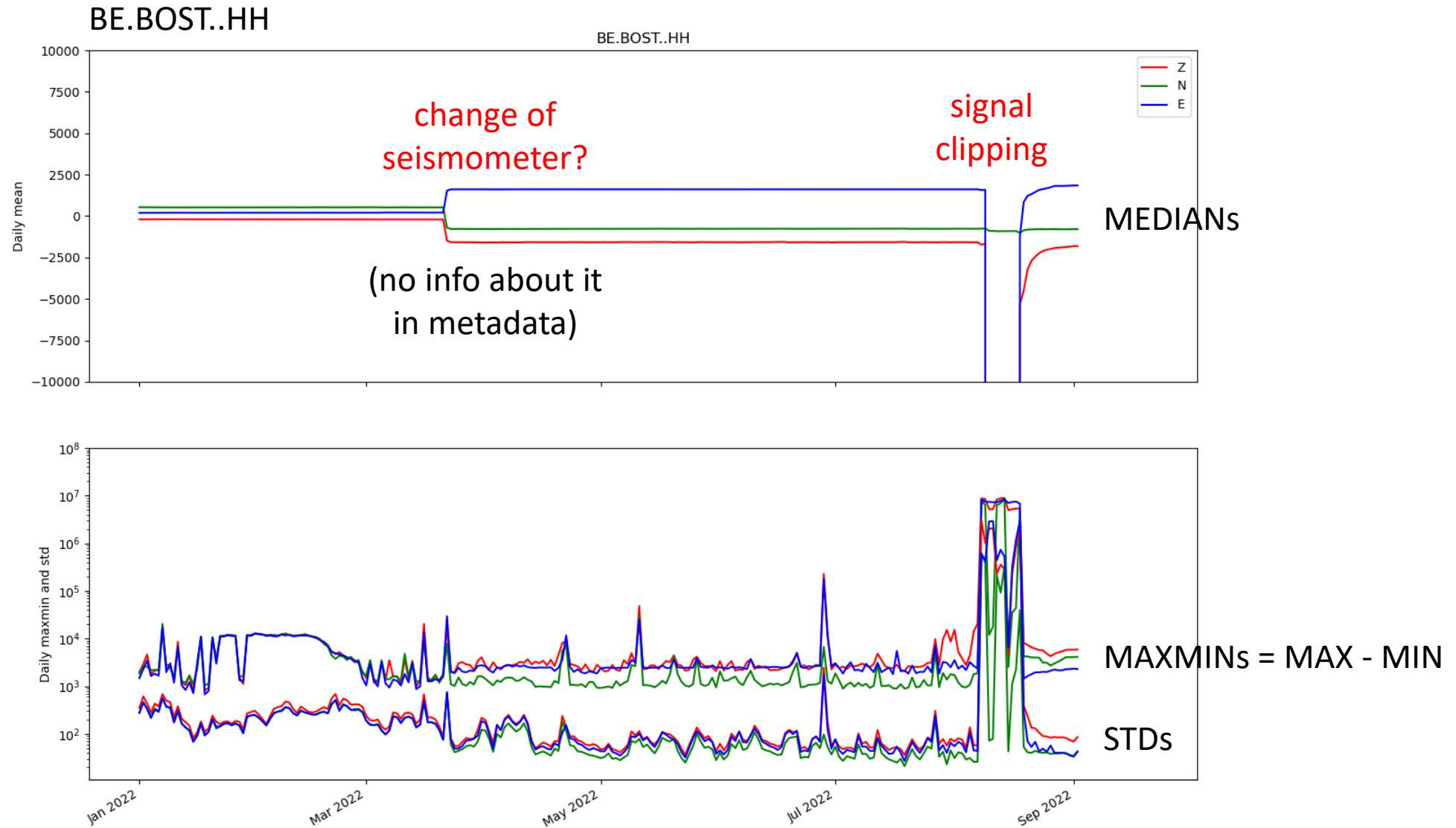
Probabilistic power spectral density (PPSD)

- time-frequency representation of the signal
- not included in EIDA WFCatalog (yet)
- need to download continuous data
- long computation of PPSD
- depends on metadata (changes/updates!)

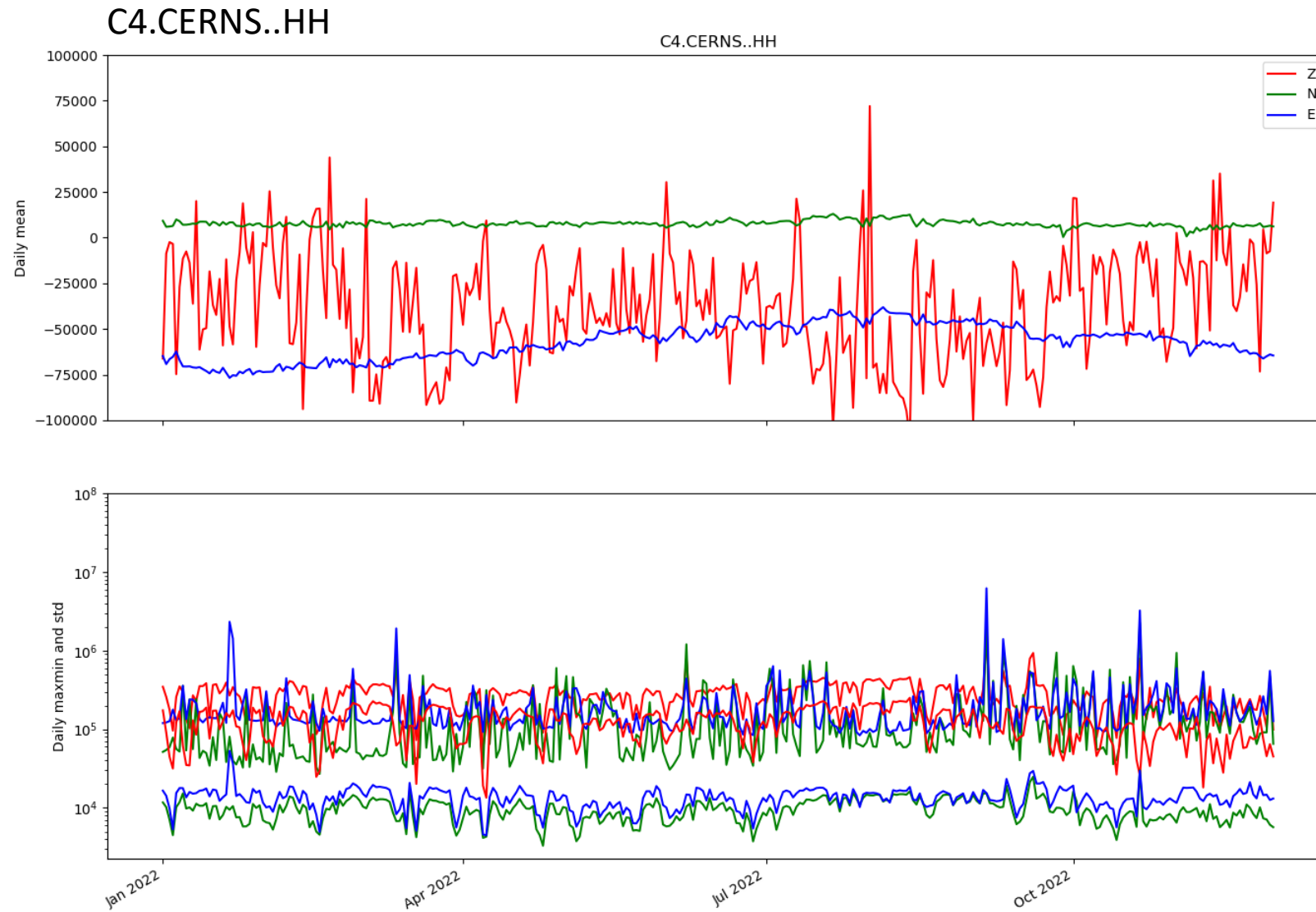
Metrics in WFCatalog

- no frequency analysis
- easy and fast to download from EIDA
- doesn't depend on metadata
- metrics:
 - daily median
 - daily std
 - daily max, min
 - ...

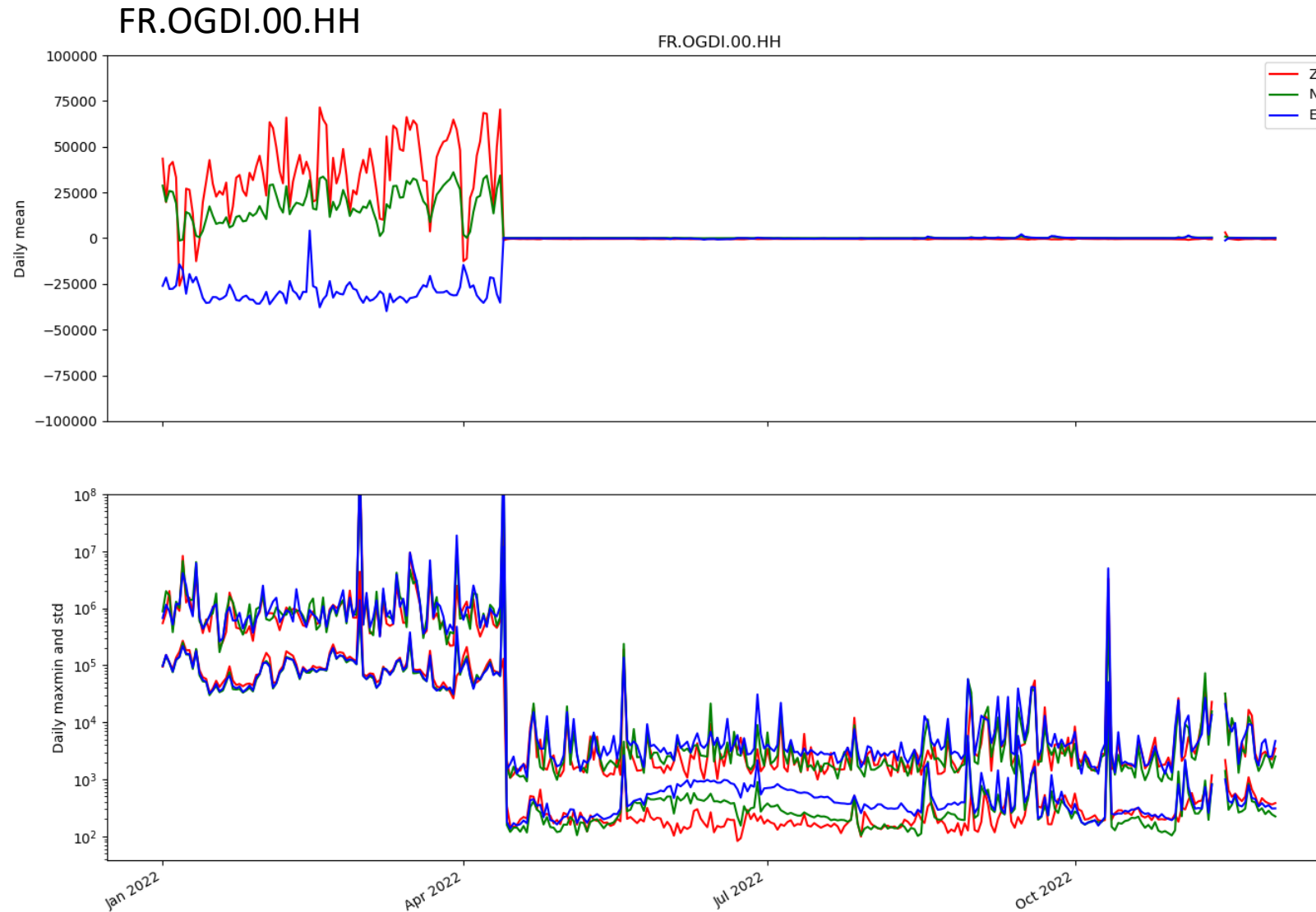
Change of seismometer, signal clipping



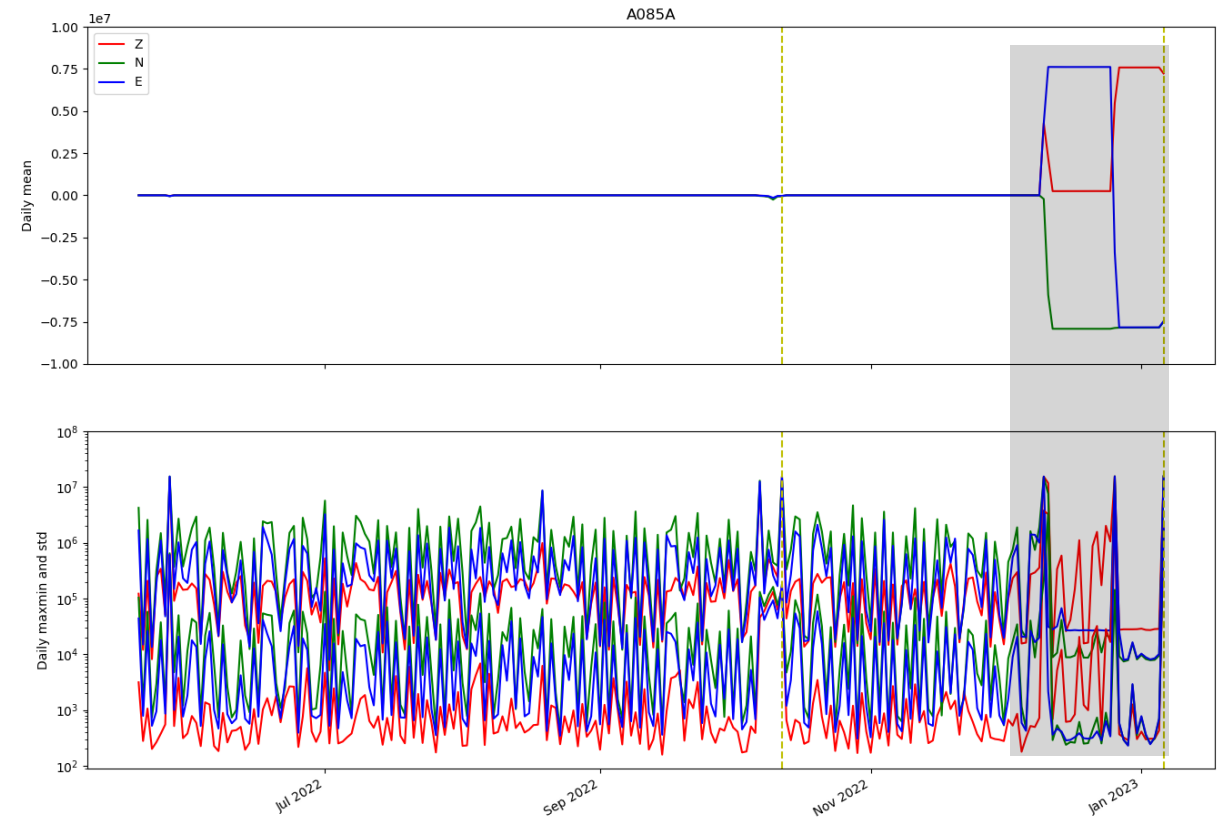
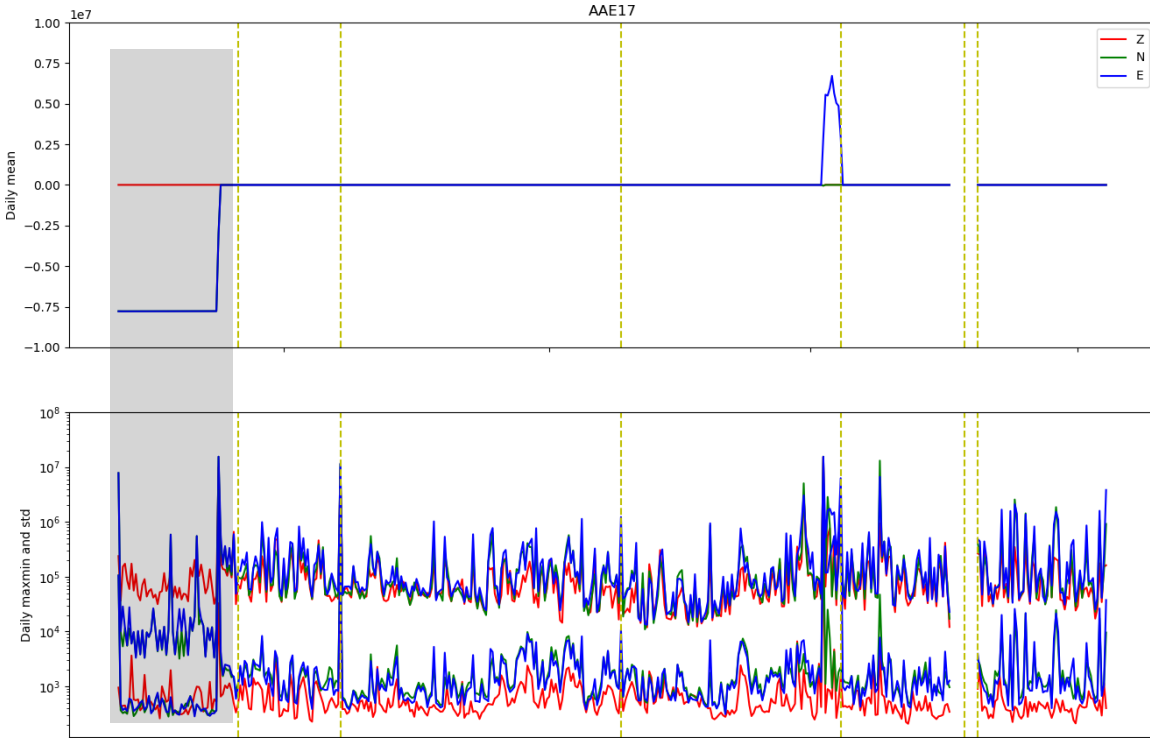
Noisy **Z** component



Improvement of station registration



Saturated signal offset, signal clipping



What is signal maximum?

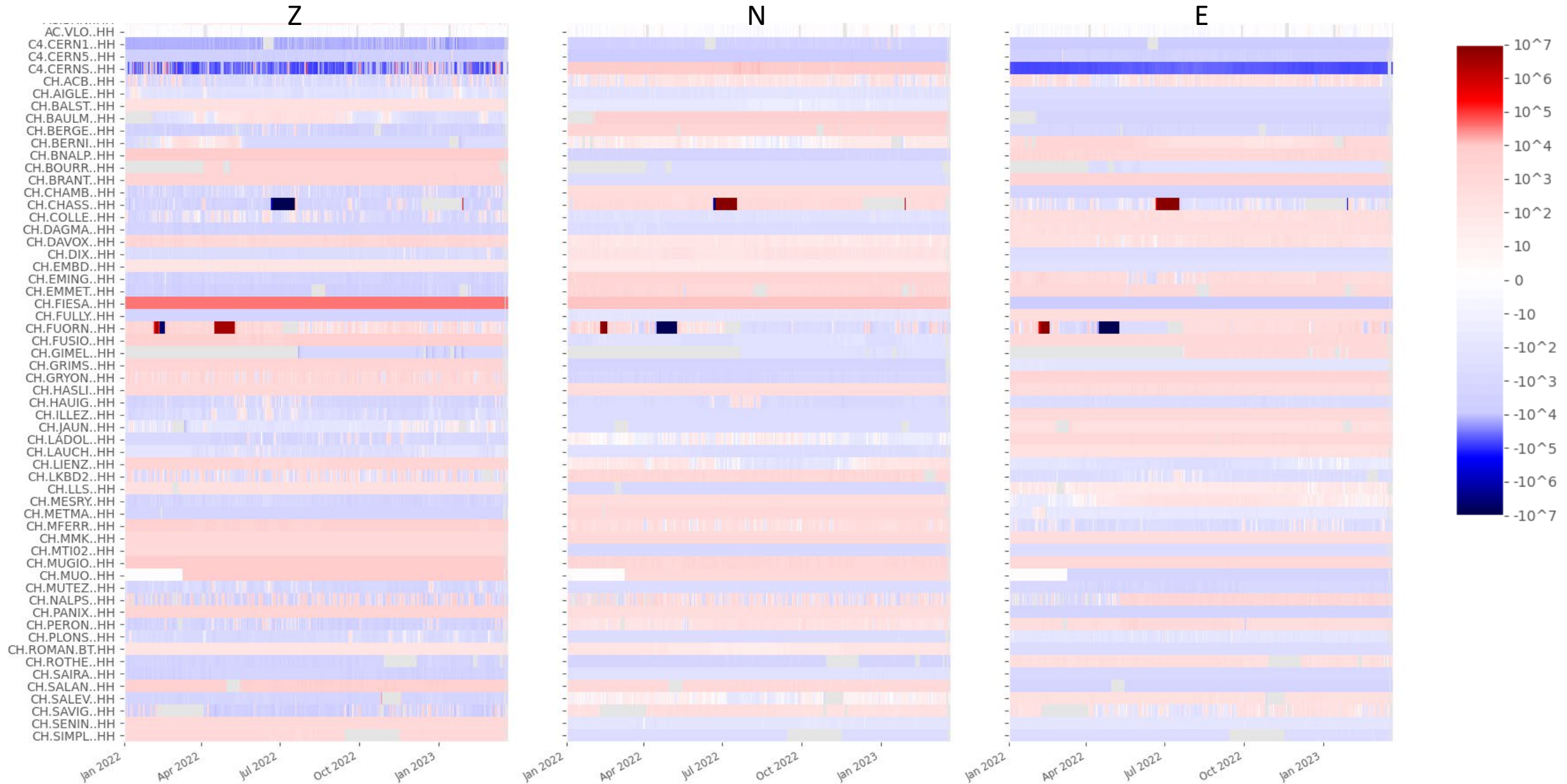
24 bit ADC = 8.4×10^6

32 bit ADC = 2×10^9

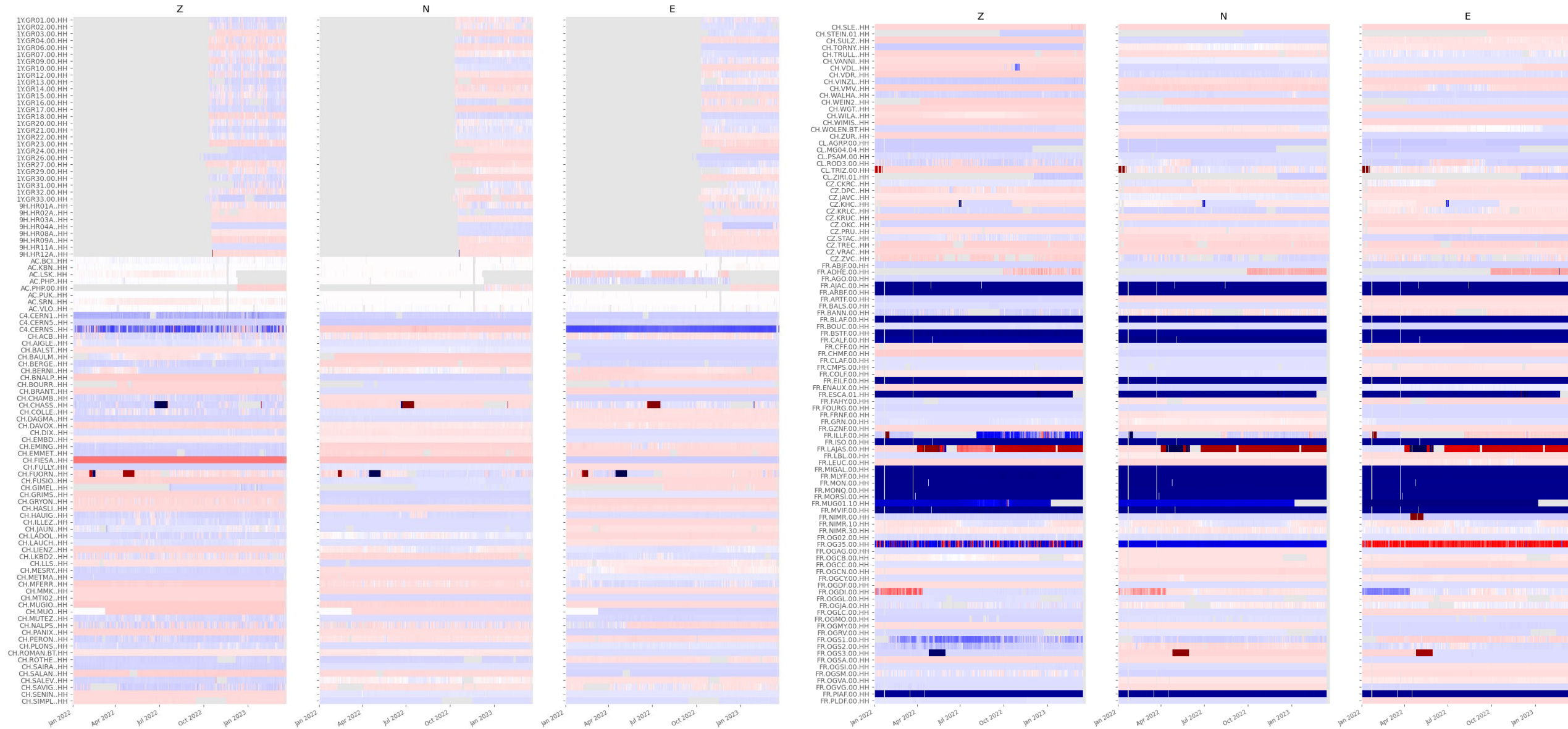
Gray areas:

saturated offsets → saturated amplitudes

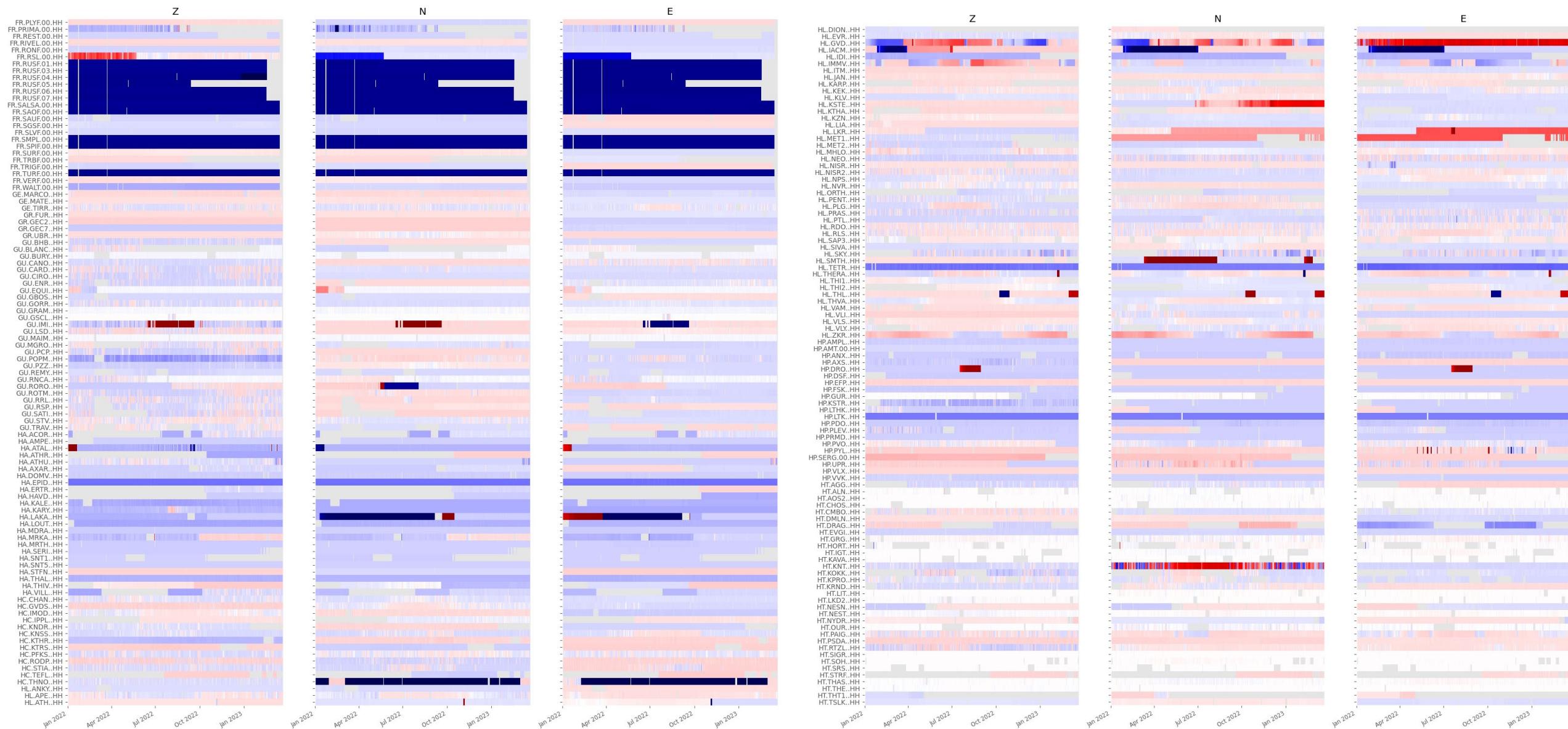
Saturation of signal daily medians/offsets



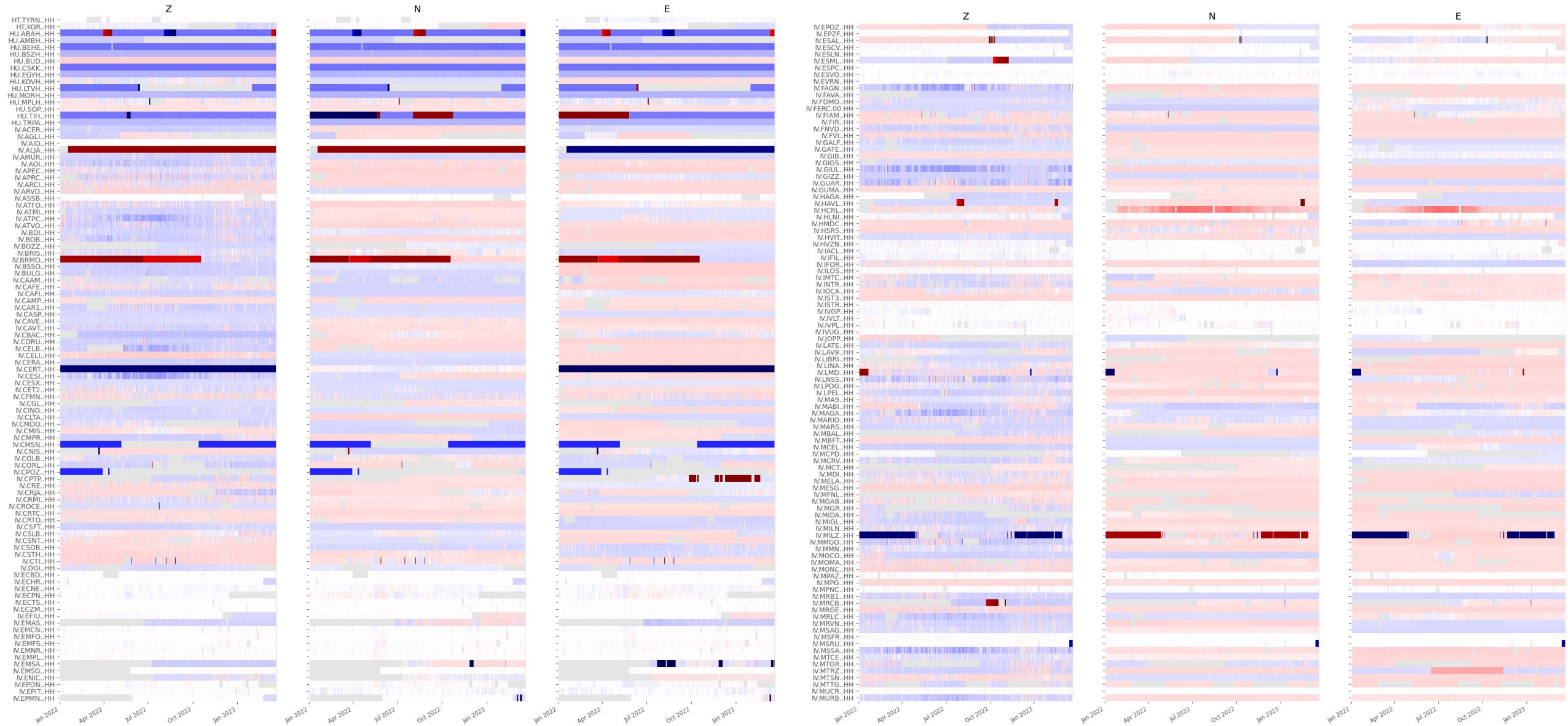
AdriaArray - daily medians #1



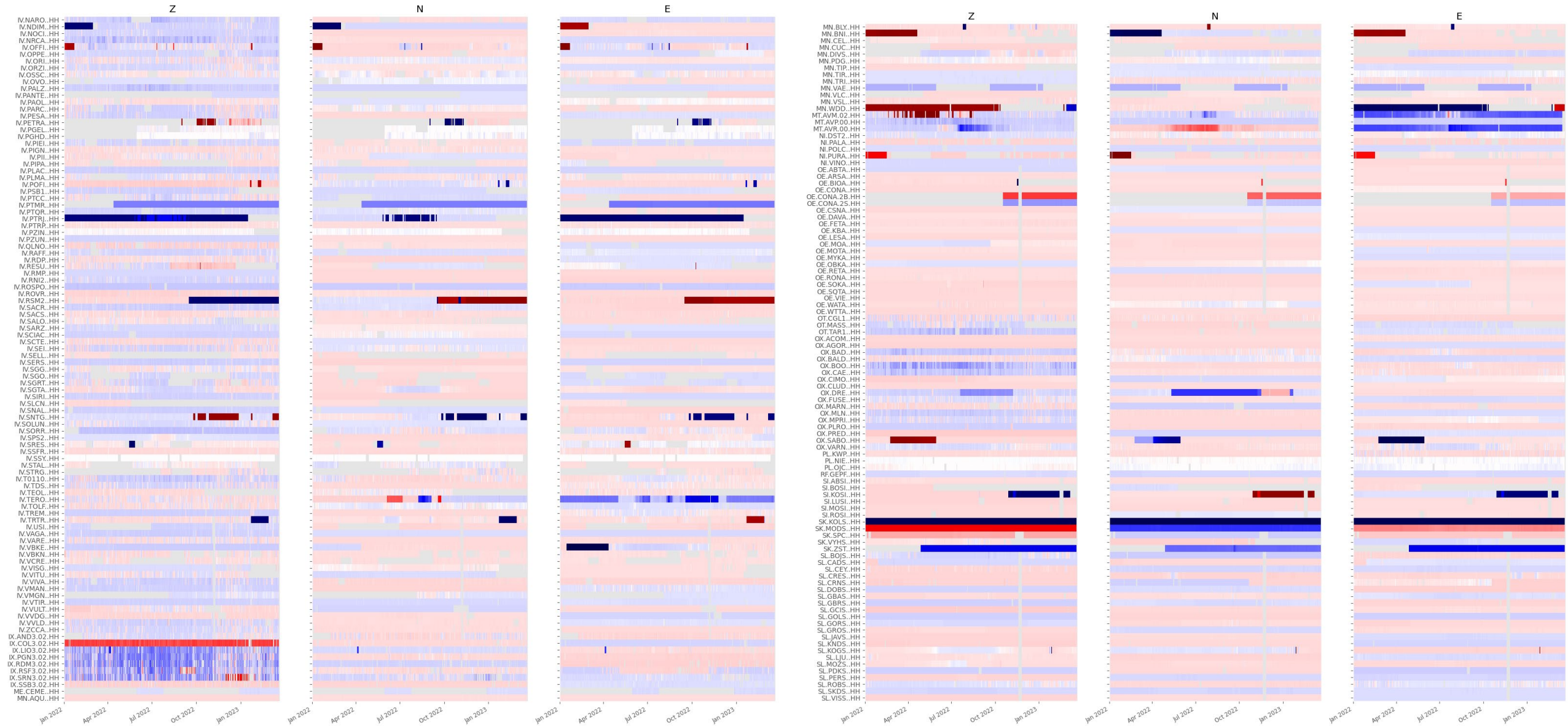
AdriaArray - daily medians #2



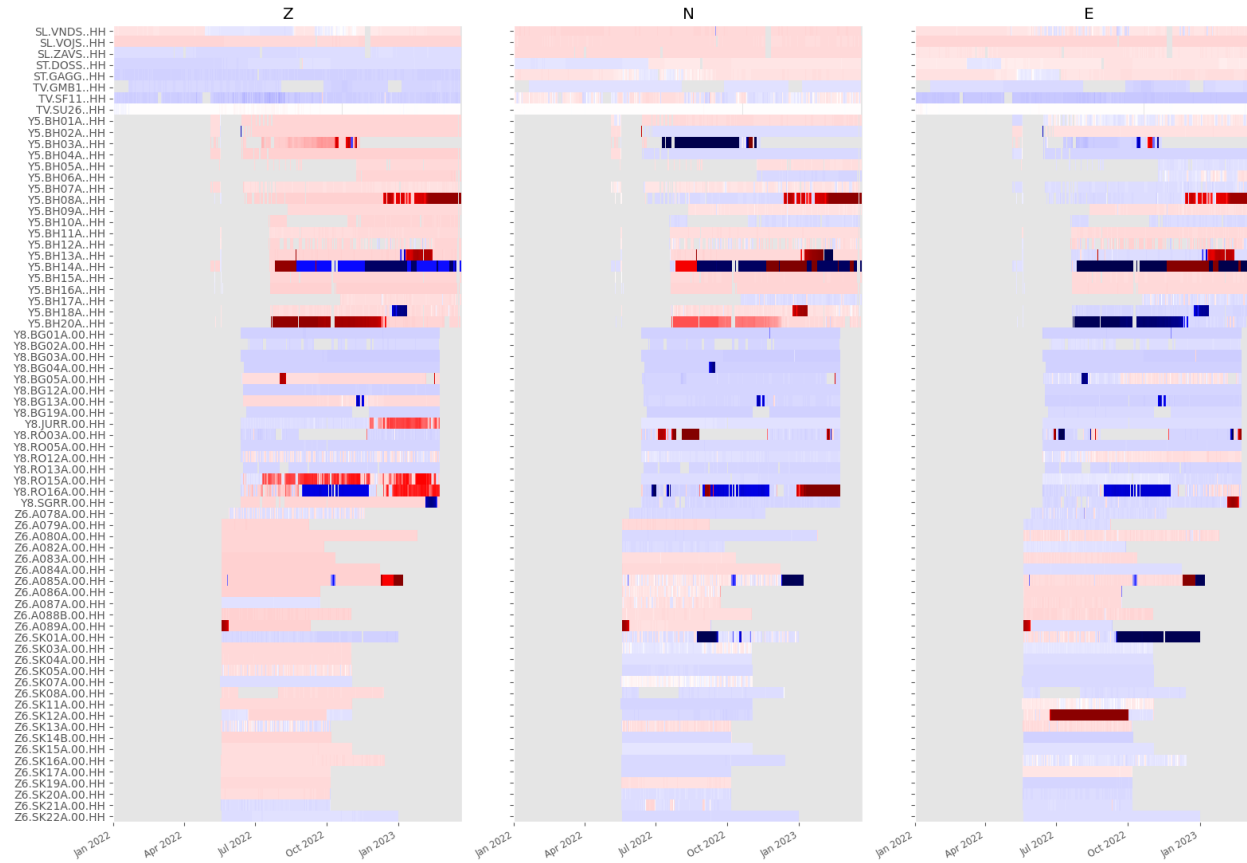
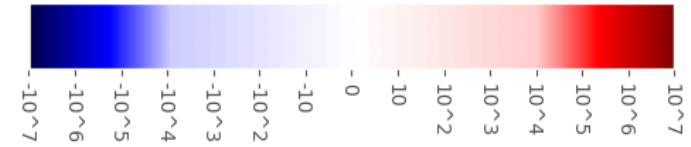
AdriaArray - daily medians #3



AdriaArray - daily medians #4



AdriaArray - daily medians #5



Saturated medians - map

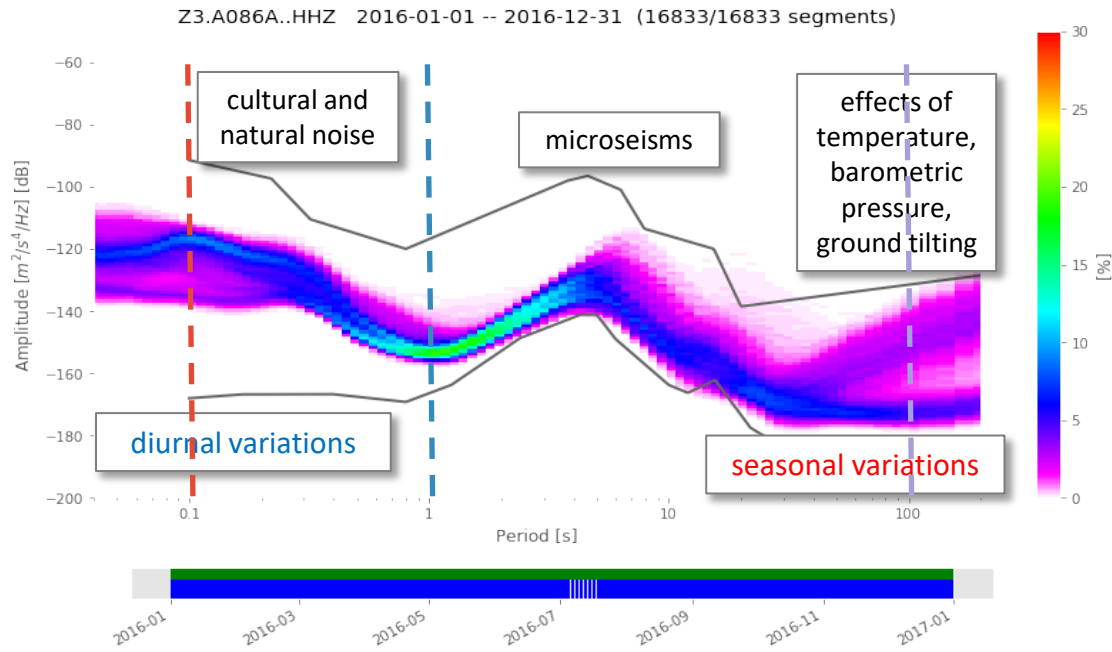
Tested period:
2022-01-01 – 2023-03-21
445 days

Saturation condition:
 $\text{abs}(\text{median}) > 1\text{e}6$

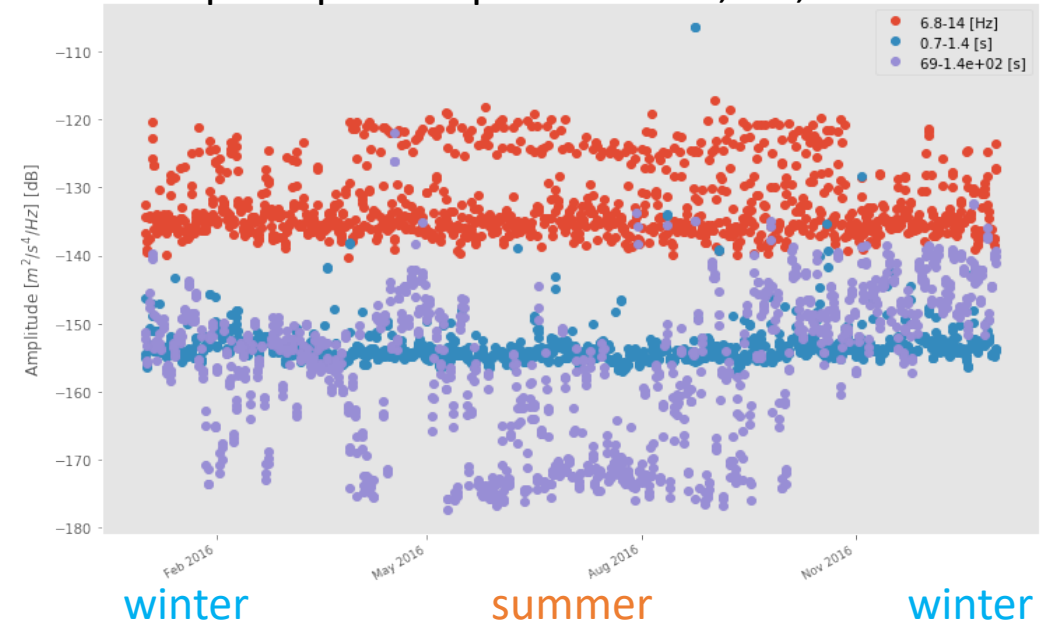


- ▲ all stations (861)
- ▲ 10+ days: 79/861 9 %
- ▲ 100+ days: 43/861 5 %
- ▲ 300+ days: 28/861 3 %

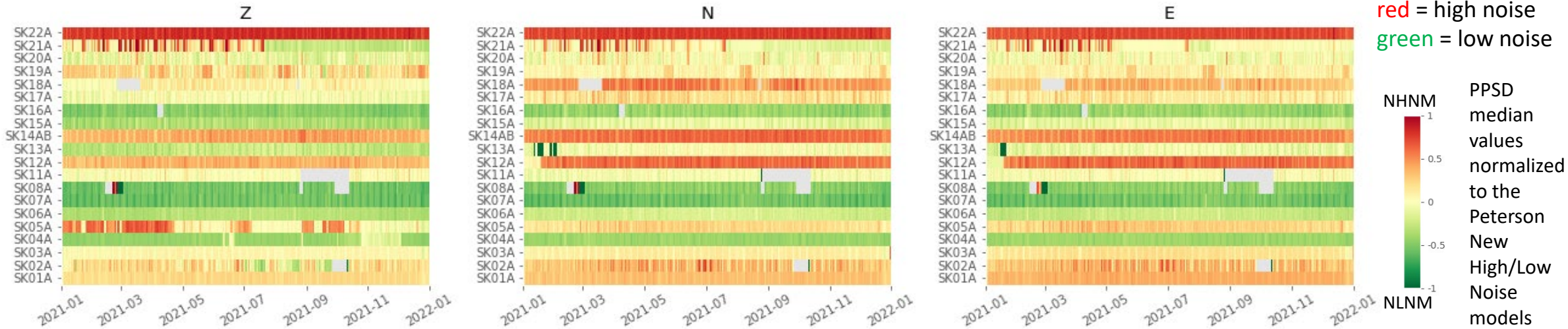
Probabilistic power spectral density (PPSD)



Temporal plot for periods 0.1 s, 1 s, 100 s

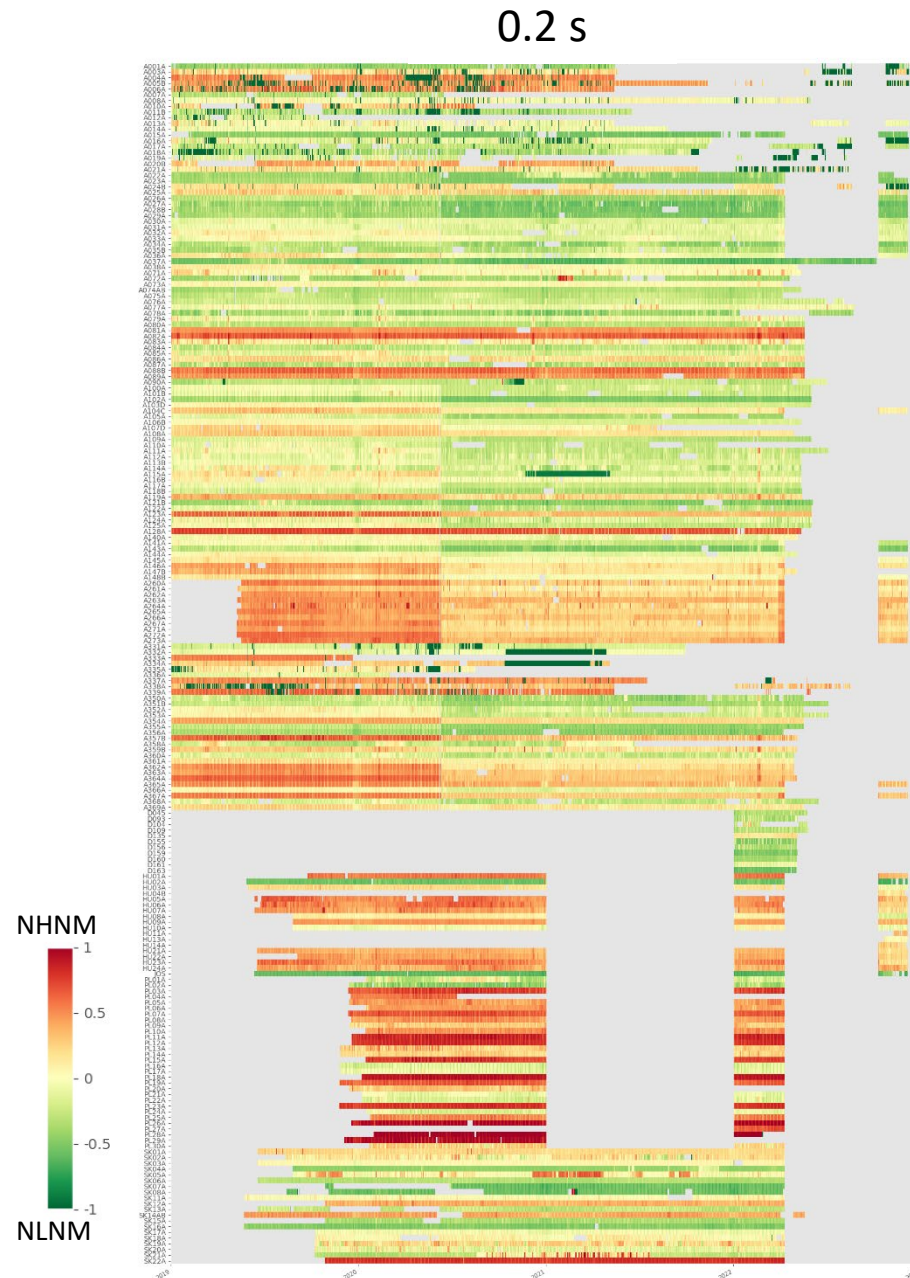


PPSDs
on
period
0.2 s



PPSDs for the PACASE experiment 2019-2022

- thank Antje Schlömer for PSD computation
- not finished yet

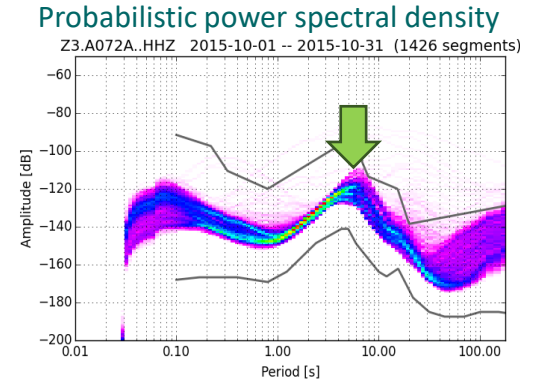
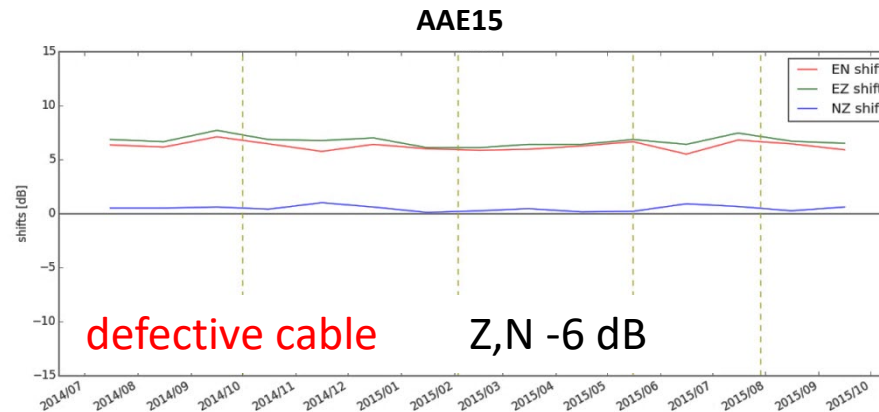
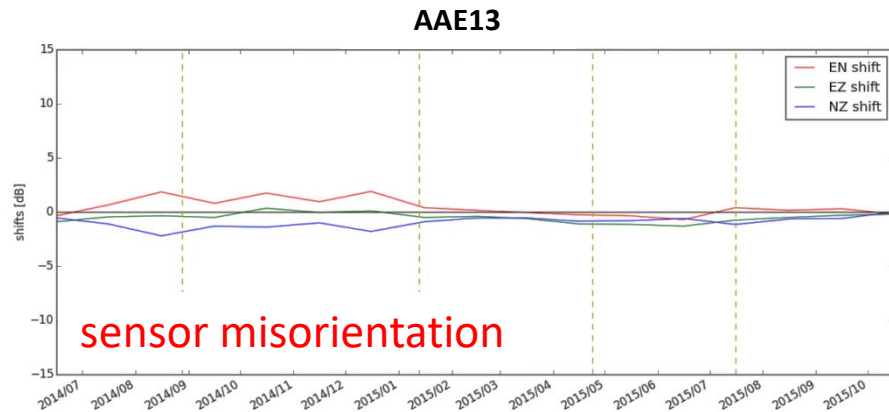
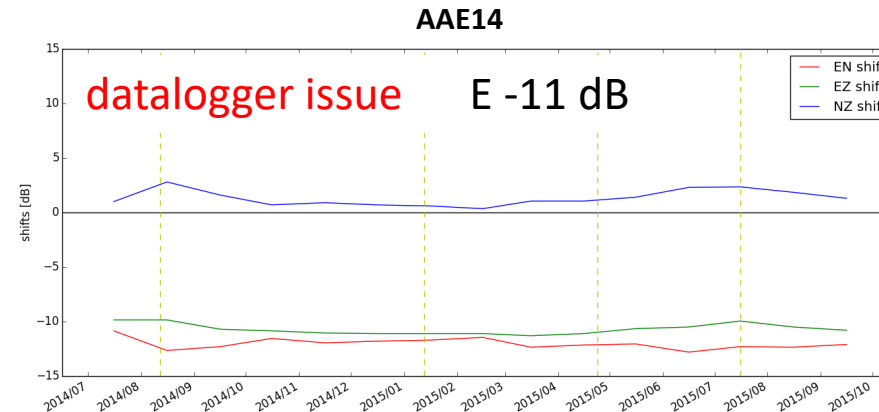
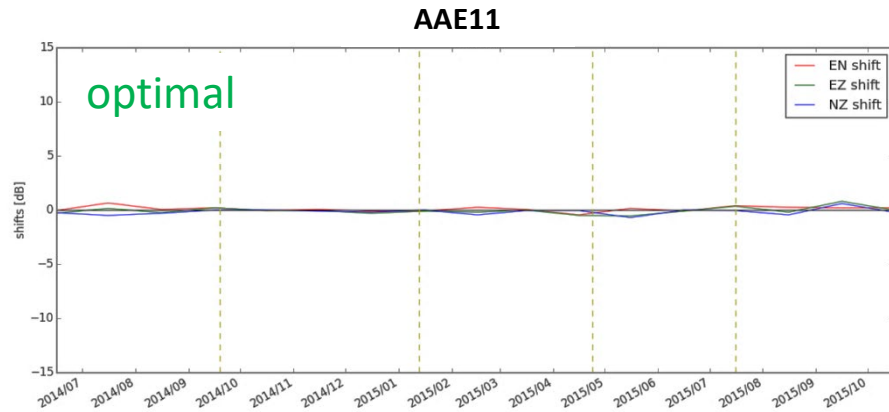


Detection of amplitude gain issues by the PPSD method

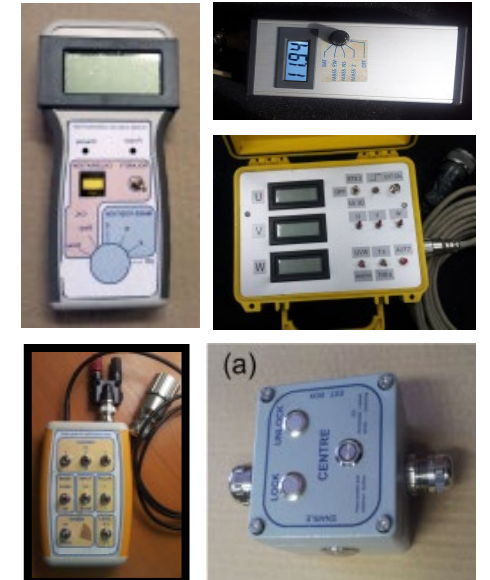
It compares ratios of PPSDs between the three components in a range of 4-8 seconds (where the secondary microseisms are substantially larger than noise from local sources).

A global and local trends of the ratios are removed.

Double check - stations with amplitude misgains are tested by special control devices for seismometers (CMG, STS) and GAIA DASS (developed in IG CAS).

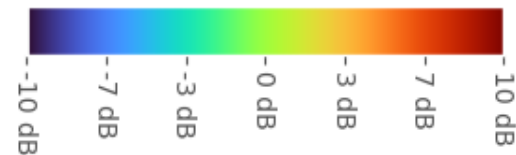
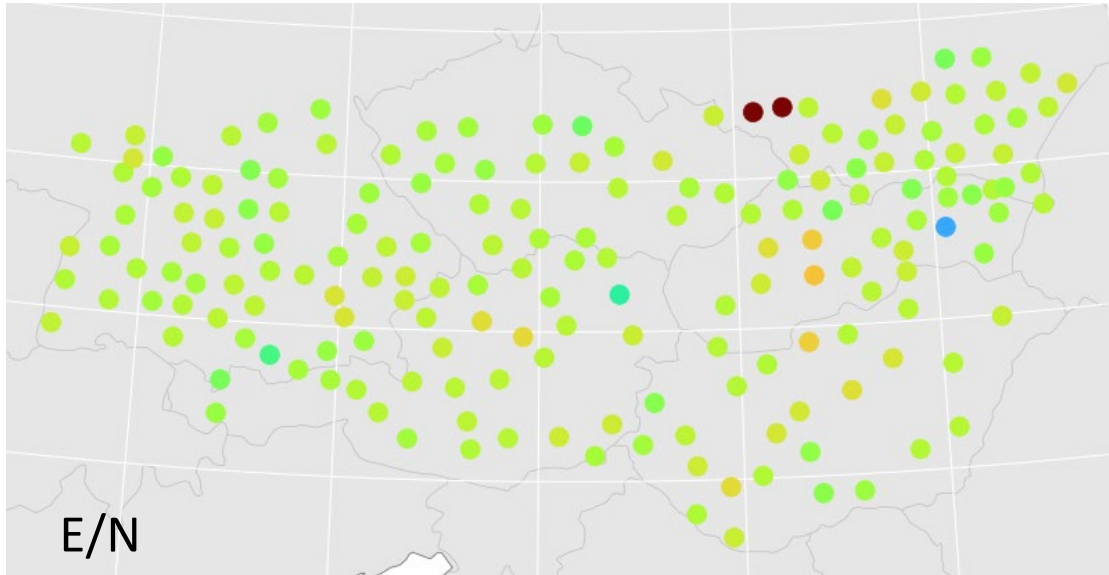


Control and calibration units

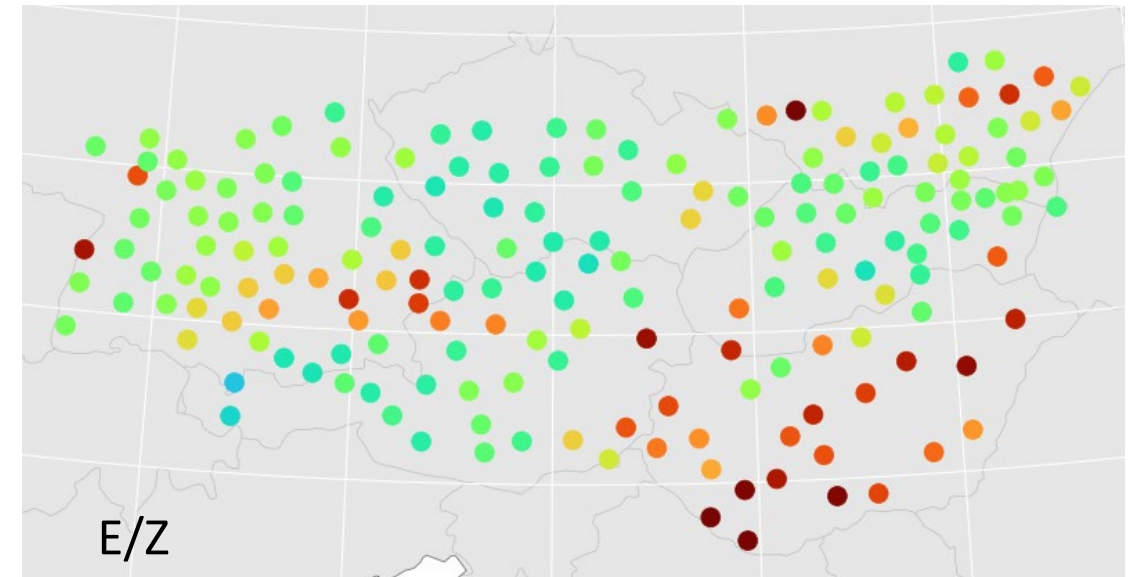
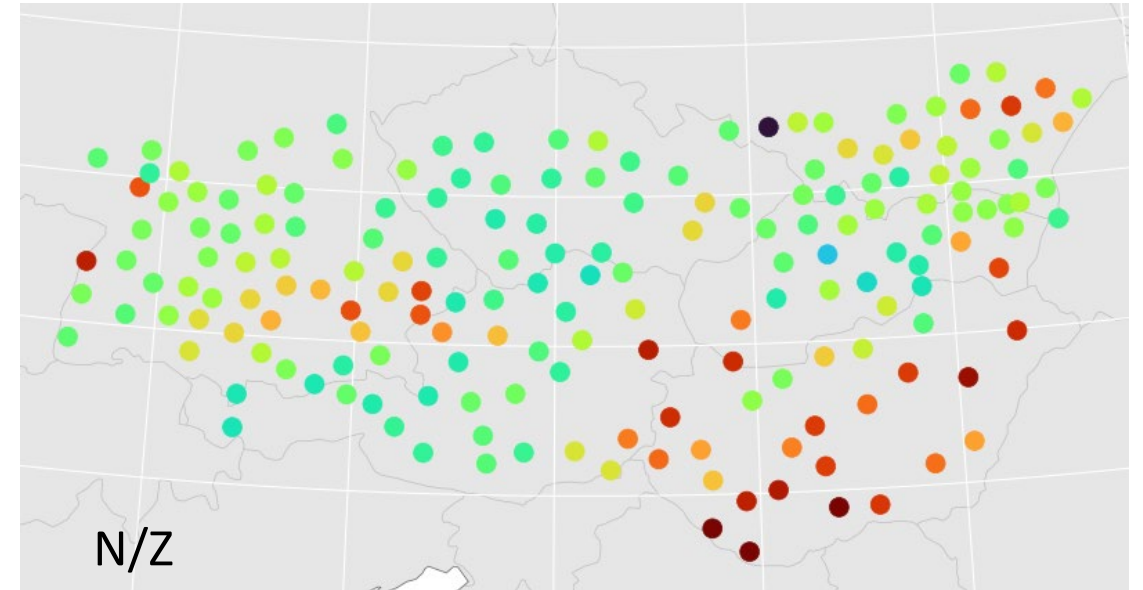


PACASE - amplitude gain issues, first results

- **global** trend subtracted

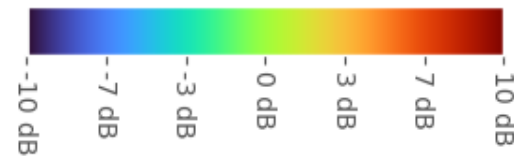
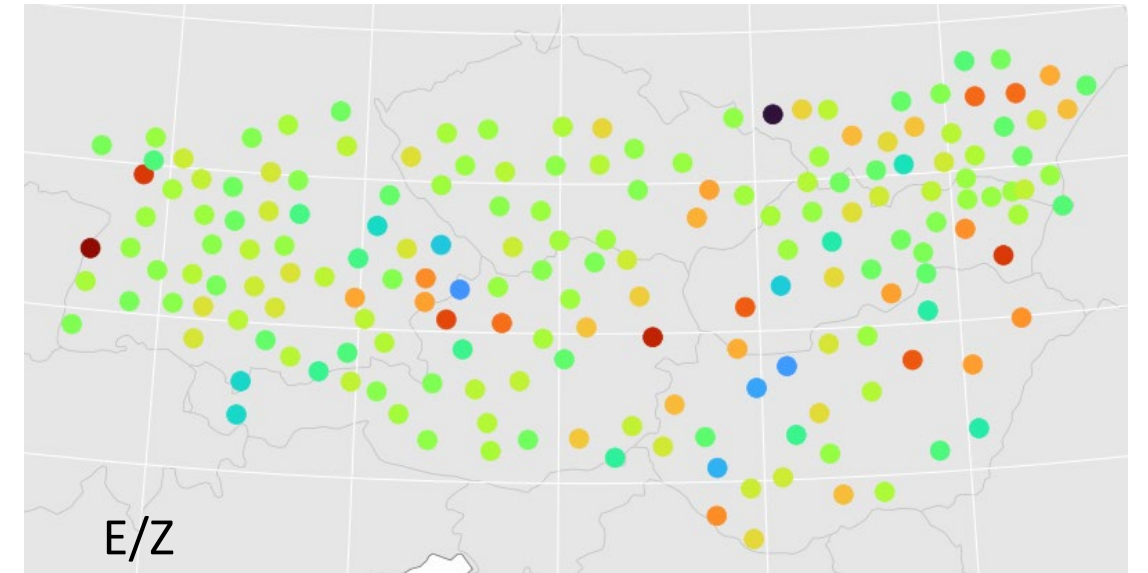
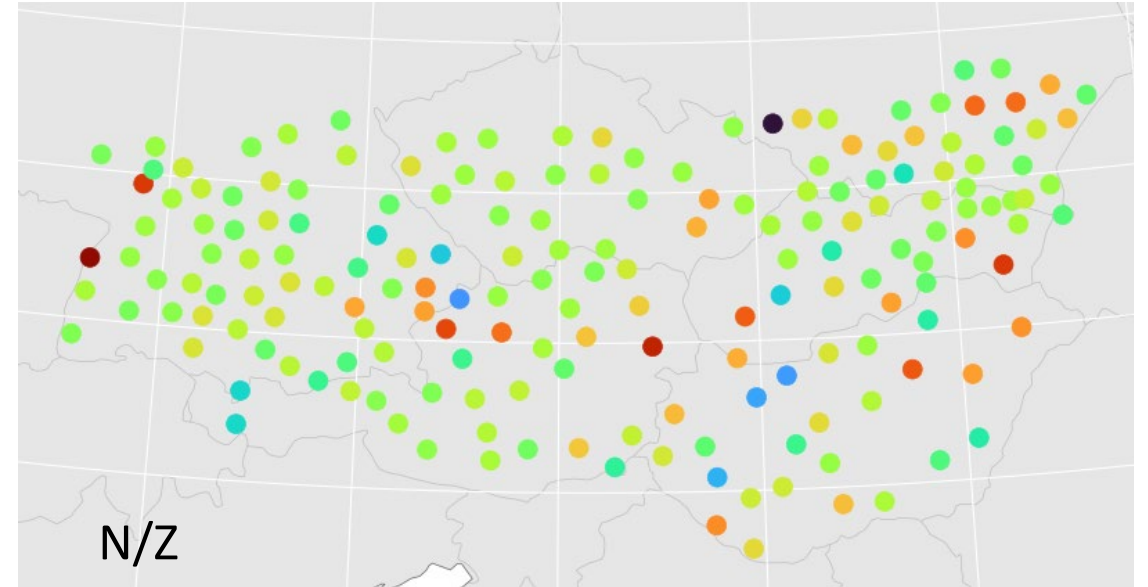
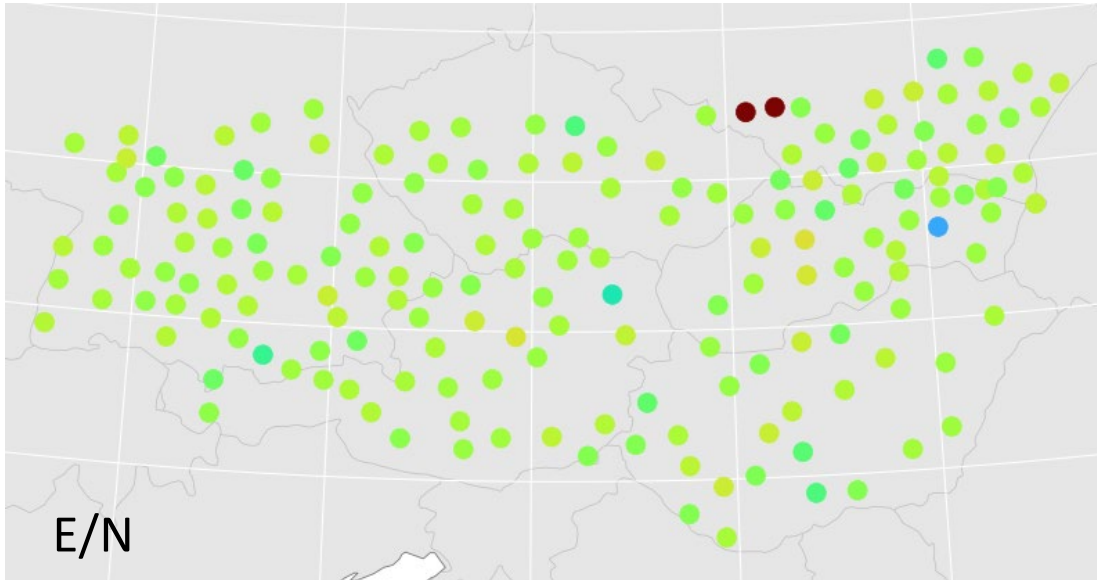


sediments?

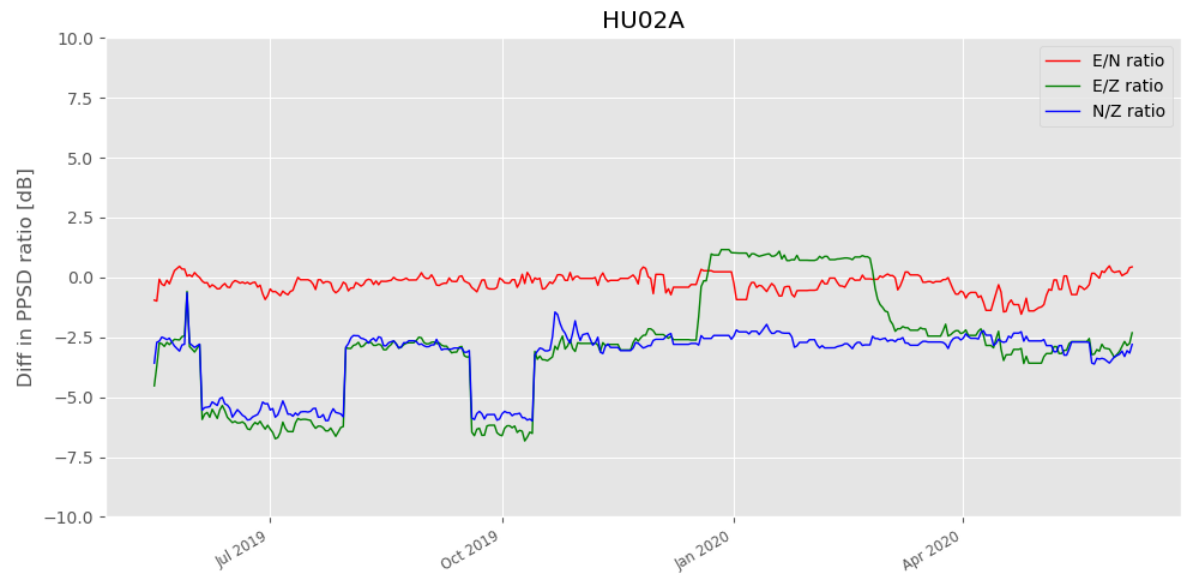
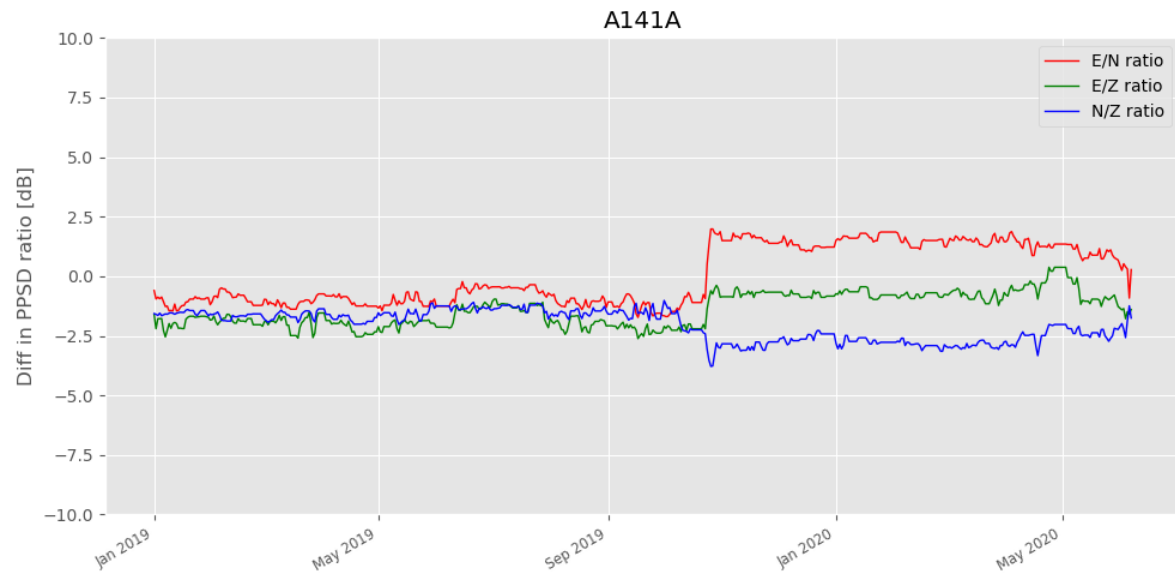
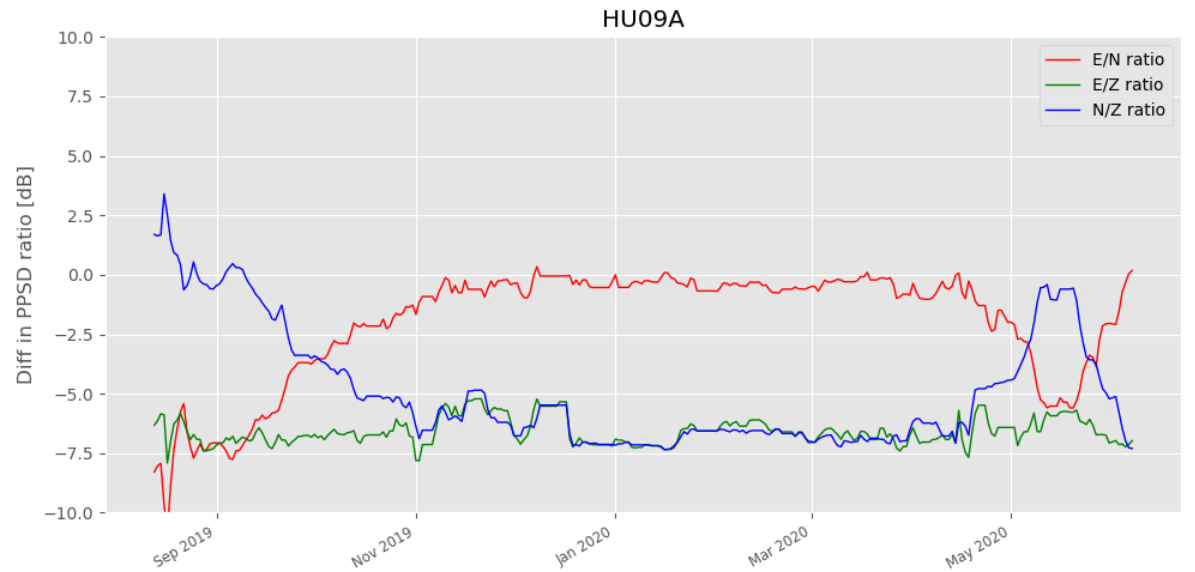
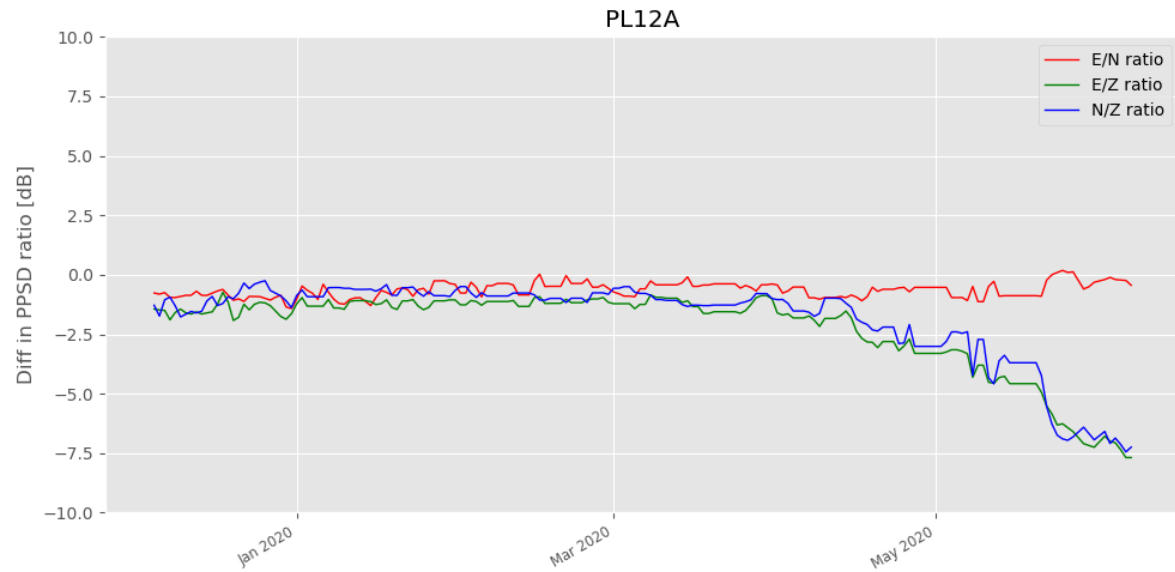


PACASE - amplitude gain issues, first results

- **local** trend subtracted



Varying amplitude misgains



Conclusions

METADATA

- lots of issues in metadata
- easy and fast to check
- slow to contact and communicate with data providers
- easy to modify
- tracker for metadata changes wanted!

PPSD

- noise level on different frequencies
- slow computation, on EIDA site?
- depends on changes in metadata

MICROSEISM PPSD

- info about amplitude misgains
(needs to complete by mis-orientation checks)
- info about sediments
- smaller dB step needed (1 dB $\rightarrow$ 0.1 dB)

DAILY MEANS, STDS, MAXMINS

- easy and fast to download from EIDA WFCatalog
- not depended on changes in metadata
- manual inspection: noisy component, change of seismometers, signal saturation/clipping, component exchange, migrating offsets, ...
- automatic detection: signal offset saturation/clipping (mean $\geq$ big), saturated/frozen signal (min $\sim$ max)

Time to regularly check data & metadata quality of AdriaArray stations?



TASK for AdA WG 3!

Thank you for your attention!

Acknowledgment

This research has been partly supported by the Grant Agency of the Czech Republic (grant no. 21-25710). The AdriaArray workshop 2023 in Dubrovnik is partly supported by ORFEUS & EPOS within the EPOS SP project (Horizon 2020 Grant Agreement No. 871121).

Several figures and calculations have been prepared with the use of the Generic Mapping Tools (Wessel and Smith, 1998) and ObsPy (Krischer et al., 2015).

We would like to thank the Seismic Network Teams of AlpArray, PACASE and AdriaArray, and the data providers of all permanent seismic networks in an AdriaArray region.

Supplements

Stations in _ADARRAY but not in Petr's tables (30.3.2023)

XP.FR04A
CH.MOUTI
CH.LASAR
CH.GOURZ
CZ.GOPC
HT.FNA
HA.PSAH
RO.BUR33
RO.PLOR4
RO.TOPG
RO.PLOR2
Y8.PL28A
CR.RABC
CR.SMRN
CR.VIRC

P 2472 None RABC -- 0
P 2475 None SMRN -- 0
P 2477 None VIRC -- 0

GE.APE
GE.GVD
GE.IMMV
GE.KARP
GE.KTHA
GE.MORC
GE.ZKR
GE.PSZ
GE.THERA
OE.JAVC

?duplicate of HL.APE
?duplicate of HL.GVD
?duplicate of HL.IMMV
?duplicate of HL.KARP
?duplicate of HL.KTHA
duplicate of CZ.MORC
?duplicate of HL.ZKR
?duplicate of HU.PSZ
?duplicate of HL.THERA
duplicate of CZ.JAVC

Not valid FDSN StationXML schemes

Station	EIDA node	scheme version	error message
AC.BERA	INGV-ONT	1.2	/FDSNStationXML/Network/Station/Channel[1]/Response/Stage[3] The content of element 'fsx:Stage' is not complete. Tag 'fsx:StageGain' expected.
AC.LACI	INGV-ONT	1.2	/FDSNStationXML/Network/Station/Channel[1]/Response/Stage[3] The content of element 'fsx:Stage' is not complete. Tag 'fsx:StageGain' expected.
OE.ABTA	ODC	1.1	/FDSNStationXML/Network/Station/Channel[1]/ClockDrift value doesn't match any pattern of ['(\\+ -)?([0-9]+(\\.([0-9]*)?)? \\.([0-9]+)([Ee](\\+ -)?[0-9]+)? INF -INF NaN']
IV.AGLI	INGV-ONT	1.2	/FDSNStationXML/Network/TotalNumberStations invalid literal for int() with base 10: "
IV.ALJA	INGV-ONT	1.2	/FDSNStationXML/Network/TotalNumberStations invalid literal for int() with base 10: "
IV.AOI	INGV-ONT	1.2	/FDSNStationXML/Network/TotalNumberStations invalid literal for int() with base 10: "
IV.APRC	INGV-ONT	1.2	/FDSNStationXML/Network/TotalNumberStations invalid literal for int() with base 10: "
IV.ARCI	INGV-ONT	1.2	/FDSNStationXML/Network/TotalNumberStations invalid literal for int() with base 10: "
IV.ATFO	INGV-ONT	1.2	/FDSNStationXML/Network/TotalNumberStations invalid literal for int() with base 10: "
...			
total 189 IV.* stations			

Metadata P&Z issues #1

easy to fix:

```
7B.A339A 0_LHE inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-0.07405+0.07405j), (-0.07405-0.07405j), (-177.72+177.72j), (-177.72+177.72j)]
7B.A339A 3_HHE inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-0.07405+0.07405j), (-0.07405-0.07405j), (-177.72+177.72j), (-177.72+177.72j)]
BW.ZUGS 2_BHZ inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-0.074024-0.07402j), (-0.074024+0.074024j), (-502.6+0j), (-1005.3+0j), (-1130.9+0j)]
HL.NISR 3_HHE inventoryP&Z: real(P)>=0 stage1: Pole=0.148597+0.148597j
HL.NISR 3_HHE inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-0.148597+0.148597j), (0.148597+0.148597j), (-1130.97+0j), (-1005.31+0j), (-502.655+0j)]
HL.NISR 4_HHN inventoryP&Z: real(P)>=0 stage1: Pole=0.148597+0.148597j
HL.NISR 4_HHN inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-0.148597+0.148597j), (0.148597+0.148597j), (-1130.97+0j), (-1005.31+0j), (-502.655+0j)]
HL.NISR 5_HHZ inventoryP&Z: real(P)>=0 stage1: Pole=0.148597+0.148597j
HL.NISR 5_HHZ inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-0.148597+0.148597j), (0.148597+0.148597j), (-1130.97+0j), (-1005.31+0j), (-502.655+0j)]
IV.TB03 2_DHZ inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-10.5529+11.7199j), (-10.553-11.72j)]
IV.TB03 11_DHZ inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-10.5529+11.7199j), (-10.553-11.72j)]
```

data providers need to fix

```
IV.TB03 0_DH1 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-10.8472+12.0467j), (-10.553-11.72j)]
IV.TB03 1_DH2 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-8.82909+9.80546j), (-10.553-11.72j)]
IV.TB03 3_DH1 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-8.61887+9.572j), (-10.553-11.72j)]
IV.TB03 4_DH2 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-7.56779+8.40468j), (-10.553-11.72j)]
IV.TB03 5_DHZ inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-7.98823+8.8716j), (-10.553-11.72j)]
IV.TB03 6_DH1 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-9.79608+10.8794j), (-10.553-11.72j)]
IV.TB03 7_DH2 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-9.58587+10.6459j), (-10.553-11.72j)]
IV.TB03 8_DHZ inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-10.721+11.9066j), (-10.553-11.72j)]
IV.TB03 9_DH1 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-10.8472+12.0467j), (-10.553-11.72j)]
IV.TB03 10_DH2 inventoryP&Z: imag(prod(Poles))!=0 stage1: Poles=[(-8.82909+9.80546j), (-10.553-11.72j)]
```


Response fails

E	AC.BERA.00.HH[ZNE]	2022-07-19T12:09:00.000000Z	None	response fail	
E	AC.LACI.00.HH[ZNE]	2022-07-19T00:00:00.000000Z	None	response fail	
E	BW.FFB2..BH1	2015-06-12T00:00:00.000000Z	None	response fail	stage2: len(Poles,Zeros)=(0,0)
E	HL.PTL..HHZ	2010-01-01T00:00:00.000000Z	2010-09-10T00:00:00.000000Z	response fail	
E	HT.CMBO..HHN	2011-02-02T00:00:00.000000Z	2013-01-01T00:00:00.000000Z	response fail	stage2: len(Poles,Zeros)=(1,0)
E	HT.KNT..HHN	2008-11-03T00:00:00.000000Z	2015-11-05T00:00:00.000000Z	response fail	stage2: len(Poles,Zeros)=(3,0)
E	IV.CNIS..HH[ZNE]	2022-03-14T23:59:59.000000Z	None	response fail	
E	IV.COLB..HH[ZNE]	2022-02-17T23:59:59.000000Z	None	response fail	
E	IV.GIUL..HH[ZNE]	2023-02-16T12:00:00.000000Z	None	response fail	
E	IV.GUAR..HH[ZNE]	2023-02-16T09:31:00.000000Z	None	response fail	
E	IV.RDP..HH[ZNE]	2023-02-14T10:00:00.000000Z	None	response fail	
E	IV.VBKE..HH[ZNE]	2021-10-09T00:00:00.000000Z	None	response fail	
E	MN.TRI..[LV]H[ZNE]	2017-08-16T10:00:00.000000Z	None	response fail	
E	OX.BAD..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	OX.BALD..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	OX.DRE..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	OX.FUSE..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	OX.MARN..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	OX.MLN..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	OX.PLRO..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	OX.PRED..[LV]H[ZNE]	2016-01-01T00:00:00.000000Z	None	response fail	
E	IV.SAMA..EH[ZNE]	2023-02-14T11:30:00.000000Z	None	response fail	
E	MT.CLP2.02.HH4	2021-04-14T00:00:00.000000Z	2021-05-14T13:59:00.000000Z	unknown component	
E	MT.CLP2.02.HH6	2021-04-14T00:00:00.000000Z	2021-05-14T13:59:00.000000Z	unknown component	
E	MT.CLP2.02.DH4	2021-05-14T14:00:00.000000Z	2500-12-31T23:59:59.000000Z	unknown component	
E	MT.CLP2.02.DH6	2021-05-14T14:00:00.000000Z	2500-12-31T23:59:59.000000Z	unknown component	
E	MT.CLP4.02.HH5	2021-04-14T00:00:00.000000Z	2021-05-14T13:59:00.000000Z	unknown component	
E	MT.CLP4.02.HH6	2021-04-14T00:00:00.000000Z	2021-05-14T13:59:00.000000Z	unknown component	
E	MT.CLP4.02.DH5	2021-05-14T14:00:00.000000Z	2500-12-31T23:59:59.000000Z	unknown component	

ObsPy warnings

```
# Input sampling rate of stage XX is inconsistent with the previous stages' output sampling rate
FR.ARBFB.00, FR.CALF.00, FR.ESCA.00, FR.OGAG.00, FR.OGDI.00, FR.RUSF.00, FR.RUSF.01, FR.SAOF.00, FR.SMPL.00, IV.CGL, IV.TB01.01, IV.TB01.18, IV.TB02.01,
IV.TB02.21, IV.TB03.01, IV.TB03.05, IV.TB03.15, IV.TB03.25, MN.AQU, MN.BLY, MN.BNI, MN.CEL, MN.CUC, MN.DIVS, MN.PDG, MN.TIP, MN.TRI, MN.TUE, MN.VLC,
MN.VSL, MN.WDD, OX.BAD, OX.BALD, OX.DRE, OX.FUSE, OX.MARN, OX.MLN, OX.PLRO, OX.PRED, OX.SABO

# OTHERS
W NI.VINO      None      None      '{http://www.fdsn.org/xml/station/1}ClockDrift' has a value of NaN. It will be skipped.
W M1.PIVO..EHZ 2018-10-18T17:00:00.000000Z None azimuth missing
W M1.RYBC..EHZ 2019-04-10T00:00:00.000000Z None azimuth missing
```

Wrong band-code naming in _ADARRAY #1

Station	Starttime	Endtime	Sampling rate	Corner period	Bandcode is/ /should be	Station	Starttime	Endtime	Sampling rate	Corner period	Bandcode is/ /should be
BS.JMB..BHZ	2004-04-01T00:00:00	-	20.0	3.0	B S	HA.AMPE..HHE	2021-09-02T08:30:00	-	100.0	1.0	H E
BS.JMB..BHN	2004-04-01T00:00:00	-	20.0	3.0	B S	HA.AMPE..HHN	2021-09-02T08:30:00	-	100.0	1.0	H E
BS.JMB..BHE	2004-04-01T00:00:00	-	20.0	3.0	B S	HA.AMPE..HHZ	2021-09-02T08:30:00	-	100.0	1.0	H E
C4.CERN1..HH3	2017-03-01T00:00:00	-	250.0	61.0	H C	HA.ATHR..HHE	2022-04-14T13:00:00	-	100.0	5.0	H E
C4.CERN1..HH2	2017-03-01T00:00:00	-	250.0	61.0	H C	HA.ATHR..HHN	2022-04-14T13:00:00	-	100.0	5.0	H E
C4.CERN1..HHZ	2017-03-01T00:00:00	-	250.0	61.0	H C	HA.ATHR..HHZ	2022-04-14T13:00:00	-	100.0	5.0	H E
C4.CERN5..HH3	2017-03-01T00:00:00	-	250.0	61.0	H C	HA.HAVD..HHE	2022-04-13T11:30:00	-	100.0	5.0	H E
C4.CERN5..HH2	2017-03-01T00:00:00	-	250.0	61.0	H C	HA.HAVD..HHN	2022-04-13T11:30:00	-	100.0	5.0	H E
C4.CERN5..HHZ	2017-03-01T00:00:00	-	250.0	61.0	H C	HA.HAVD..HHZ	2022-04-13T11:30:00	-	100.0	5.0	H E
C4.CERNS..HHE	2017-02-15T00:00:00	-	250.0	61.0	H C	HA.MRTH..HHE	2020-09-08T12:00:00	-	100.0	5.0	H E
C4.CERNS..HHN	2017-02-15T00:00:00	-	250.0	61.0	H C	HA.MRTH..HHN	2020-09-08T12:00:00	-	100.0	5.0	H E
C4.CERNS..HHZ	2017-02-15T00:00:00	-	250.0	61.0	H C	HA.MRTH..HHZ	2020-09-08T12:00:00	-	100.0	5.0	H E
CZ.KHC..SHE	2003-10-27T00:00:00	-	20.0	123.0	S B	HA.SNT5..HHE	2021-06-29T08:00:00	-	100.0	1.0	H E
CZ.KHC..SHN	2003-10-27T00:00:00	-	20.0	123.0	S B	HA.SNT5..HHN	2021-06-29T08:00:00	-	100.0	1.0	H E
CZ.KHC..SHZ	2003-10-27T00:00:00	-	20.0	123.0	S B	HA.SNT5..HHZ	2021-06-29T08:00:00	-	100.0	1.0	H E
CZ.KHC..EHE	2010-05-07T00:00:00	-	100.0	123.0	E H	HA.STFN..HHE	2019-07-20T14:00:00	-	100.0	1.0	H E
CZ.KHC..EHN	2010-05-07T00:00:00	-	100.0	123.0	E H	HA.STFN..HHN	2019-07-20T14:00:00	-	100.0	1.0	H E
CZ.KHC..EHZ	2010-05-07T00:00:00	-	100.0	123.0	E H	HA.STFN..HHZ	2019-07-20T14:00:00	-	100.0	1.0	H E
CZ.PRU..SHE	2003-08-11T00:00:00	-	20.0	123.0	S B	HA.THIV..HHE	2021-08-03T09:30:00	2022-09-20T08:59:59	100.0	1.0	H E
CZ.PRU..SHN	2003-08-11T00:00:00	-	20.0	123.0	S B	HA.THIV..HHN	2021-08-03T09:30:00	2022-09-20T08:59:59	100.0	1.0	H E
CZ.PRU..SHZ	2003-08-11T00:00:00	-	20.0	123.0	S B	HA.THIV..HHZ	2021-08-03T09:30:00	2022-09-20T08:59:59	100.0	1.0	H E
CZ.PRU..EHE	2010-05-26T00:00:00	-	100.0	123.0	E H	HC.TEFL..HHE	2022-07-13T00:00:00	-	100.0	5.0	H E
CZ.PRU..EHN	2010-05-26T00:00:00	-	100.0	123.0	E H	HC.TEFL..HHN	2022-07-13T00:00:00	-	100.0	5.0	H E
CZ.PRU..EHZ	2010-05-26T00:00:00	-	100.0	123.0	E H	HC.TEFL..HHZ	2022-07-13T00:00:00	-	100.0	5.0	H E
GE.MATE..SHZ	2007-04-24T00:00:00	-	50.0	123.0	S B	HT.DMLN..HHE	2014-09-19T00:00:00	-	100.0	1.0	H E
GE.MATE..SHE	2007-04-24T00:00:00	-	50.0	123.0	S B	HT.DMLN..HHN	2014-09-19T00:00:00	-	100.0	1.0	H E
GE.MATE..SHN	2007-04-24T00:00:00	-	50.0	123.0	S B	HT.DMLN..HHZ	2014-09-19T00:00:00	-	100.0	1.0	H E
GE.TIRR..SHZ	2003-10-13T00:00:00	-	50.0	123.0	S B	HT.NESN..HHE	2021-03-06T09:00:00	-	100.0	1.0	H E
GE.TIRR..SHE	2003-10-13T00:00:00	-	50.0	123.0	S B	HT.NESN..HHN	2021-03-06T09:00:00	-	100.0	1.0	H E
GE.TIRR..SHN	2003-10-13T00:00:00	-	50.0	123.0	S B	HT.NESN..HHZ	2021-03-06T09:00:00	-	100.0	1.0	H E
GU.MAIM..BHE	2018-12-19T10:01:00	-	100.0	41.0	B H	HT.NYDR..HHE	2016-04-01T00:00:00	-	100.0	1.0	H E
GU.MAIM..BHN	2018-12-19T10:01:00	-	100.0	41.0	B H	HT.NYDR..HHN	2016-04-01T00:00:00	-	100.0	1.0	H E
GU.MAIM..BHZ	2018-12-19T10:01:00	-	100.0	41.0	B H	HT.NYDR..HHZ	2016-04-01T00:00:00	-	100.0	1.0	H E
GU.MAIM..LHE	2018-12-19T10:01:00	-	100.0	41.0	L H	HT.TSLK..HHE	2012-10-10T00:00:00	-	100.0	1.0	H E
GU.MAIM..LHN	2018-12-19T10:01:00	-	100.0	41.0	L H	HT.TSLK..HHN	2012-10-10T00:00:00	-	100.0	1.0	H E
GU.MAIM..LHZ	2018-12-19T10:01:00	-	100.0	41.0	L H	HT.TSLK..HHZ	2012-10-10T00:00:00	-	100.0	1.0	H E

Wrong band-code naming in _ADARRAY #2

Station	Starttime	Endtime	Sampling rate	Corner period	Bandcode is/ /should be	Station	Starttime	Endtime	Sampling rate	Corner period	Bandcode is/ /should be
HU.PSZ..SHZ	2017-07-05T10:00:00	-	50.0	123.0	S B	IV.RAFF..BHE	2023-03-20T11:39:00	-	100.0	123.0	B H
HU.PSZ..SHN	2017-07-05T10:00:00	-	50.0	123.0	S B	IV.RAFF..BHN	2023-03-20T11:39:00	-	100.0	123.0	B H
HU.PSZ..SHE	2017-07-05T10:00:00	-	50.0	123.0	S B	IV.RAFF..BHZ	2023-03-20T11:39:00	-	100.0	123.0	B H
IV.CAFI..BHE	2021-06-10T10:30:00	-	100.0	41.0	B H	IV.RAFF..LHE	2023-03-20T11:39:00	-	100.0	123.0	L H
IV.CAFI..BHN	2021-06-10T10:30:00	-	100.0	41.0	B H	IV.RAFF..LHN	2023-03-20T11:39:00	-	100.0	123.0	L H
IV.CAFI..BHZ	2021-06-10T10:30:00	-	100.0	41.0	B H	IV.RAFF..LHZ	2023-03-20T11:39:00	-	100.0	123.0	L H
IV.CAFI..LHE	2021-06-10T10:30:00	-	100.0	41.0	L H	IV.RESU..BHE	2023-03-20T12:22:00	-	100.0	123.0	B H
IV.CAFI..LHN	2021-06-10T10:30:00	-	100.0	41.0	L H	IV.RESU..BHN	2023-03-20T12:22:00	-	100.0	123.0	B H
IV.CAFI..LHZ	2021-06-10T10:30:00	-	100.0	41.0	L H	IV.RESU..BHZ	2023-03-20T12:22:00	-	100.0	123.0	B H
IV.CMDO..BHE	2023-01-10T05:05:00	-	100.0	123.0	B H	IV.RESU..LHE	2023-03-20T12:22:00	-	100.0	123.0	L H
IV.CMDO..BHN	2023-01-10T05:05:00	-	100.0	123.0	B H	IV.RESU..LHN	2023-03-20T12:22:00	-	100.0	123.0	L H
IV.CMDO..BHZ	2023-01-10T05:05:00	-	100.0	123.0	B H	IV.RESU..LHZ	2023-03-20T12:22:00	-	100.0	123.0	L H
IV.CMDO..LHE	2023-01-10T05:05:00	-	100.0	123.0	L H	MN.AQU..VHE	2019-03-01T00:00:00	-	1.0	128.0	V L
IV.CMDO..LHN	2023-01-10T05:05:00	-	100.0	123.0	L H	MN.AQU..VHN	2019-03-01T00:00:00	-	1.0	128.0	V L
IV.CMDO..LHZ	2023-01-10T05:05:00	-	100.0	123.0	L H	MN.AQU..VHZ	2019-03-01T00:00:00	-	1.0	128.0	V L
IV.CORL..BHE	2022-12-02T13:30:00	-	100.0	123.0	B H	MN.BLY..VHE	2017-12-13T12:00:00	-	1.0	128.0	V L
IV.CORL..BHN	2022-12-02T13:30:00	-	100.0	123.0	B H	MN.BLY..VHN	2017-12-13T12:00:00	-	1.0	128.0	V L
IV.CORL..BHZ	2022-12-02T13:30:00	-	100.0	123.0	B H	MN.BLY..VHZ	2017-12-13T12:00:00	-	1.0	128.0	V L
IV.CORL..LHE	2022-12-02T13:30:00	-	100.0	123.0	L H	MN.BNI..VHE	2016-10-26T09:55:00	-	1.0	128.0	V L
IV.CORL..LHN	2022-12-02T13:30:00	-	100.0	123.0	L H	MN.BNI..VHN	2016-10-26T09:55:00	-	1.0	128.0	V L
IV.CORL..LHZ	2022-12-02T13:30:00	-	100.0	123.0	L H	MN.BNI..VHZ	2016-10-26T09:55:00	-	1.0	128.0	V L
IV.FERC.00.HHE	2021-09-14T09:56:00	-	100.0	0.0	H E	MN.CEL..VHE	2017-10-05T10:00:00	-	1.0	128.0	V L
IV.FERC.00.HHN	2021-09-14T09:56:00	-	100.0	0.0	H E	MN.CEL..VHN	2017-10-05T10:00:00	-	1.0	128.0	V L
IV.FERC.00.HHZ	2021-09-14T09:56:00	-	100.0	0.0	H E	MN.CEL..VHZ	2017-10-05T10:00:00	-	1.0	128.0	V L
IV.GALF..BHE	2023-03-20T21:00:00	-	100.0	123.0	B H	MN.CUC..VHE	2017-09-30T12:00:00	-	1.0	128.0	V L
IV.GALF..BHN	2023-03-20T21:00:00	-	100.0	123.0	B H	MN.CUC..VHN	2017-09-30T12:00:00	-	1.0	128.0	V L
IV.GALF..BHZ	2023-03-20T21:00:00	-	100.0	123.0	B H	MN.CUC..VHZ	2017-09-30T12:00:00	-	1.0	128.0	V L
IV.GALF..LHE	2023-03-20T21:00:00	-	100.0	123.0	L H	MN.DIVS..VHE	2022-04-08T00:00:00	-	1.0	362.0	V L
IV.GALF..LHN	2023-03-20T21:00:00	-	100.0	123.0	L H	MN.DIVS..VHN	2022-04-08T00:00:00	-	1.0	362.0	V L
IV.GALF..LHZ	2023-03-20T21:00:00	-	100.0	123.0	L H	MN.DIVS..VHZ	2022-04-08T00:00:00	-	1.0	362.0	V L
IV.PETRA..BHE	2022-11-22T00:00:00	-	100.0	123.0	B H	MN.PDG..VHE	2018-04-23T08:00:00	-	1.0	128.0	V L
IV.PETRA..BHN	2022-11-22T00:00:00	-	100.0	123.0	B H	MN.PDG..VHN	2018-04-23T08:00:00	-	1.0	128.0	V L
IV.PETRA..BHZ	2022-11-22T00:00:00	-	100.0	123.0	B H	MN.PDG..VHZ	2018-04-23T08:00:00	-	1.0	128.0	V L
IV.PETRA..LHE	2022-11-22T00:00:00	-	100.0	123.0	L H	MN.TIP..VHE	2017-10-02T18:00:00	-	1.0	128.0	V L
IV.PETRA..LHN	2022-11-22T00:00:00	-	100.0	123.0	L H	MN.TIP..VHN	2017-10-02T18:00:00	-	1.0	128.0	V L
IV.PETRA..LHZ	2022-11-22T00:00:00	-	100.0	123.0	L H	MN.TIP..VHZ	2017-10-02T18:00:00	-	1.0	128.0	V L

Wrong band-code naming in _ADARRAY #3

Station	Starttime	Endtime	Sampling rate [Hz]	Corner period [s]	Bandcode is/ /should be
MN.TUE..VHE	2017-07-26T09:02:00	-	1.0	128.0	V L
MN.TUE..VHN	2017-07-26T09:02:00	-	1.0	128.0	V L
MN.TUE..VHZ	2017-07-26T09:02:00	-	1.0	128.0	V L
MN.VLC..VHE	2022-05-03T13:09:00	2099-05-03T14:09:00	1.0	128.0	V L
MN.VLC..VHN	2022-05-03T13:09:00	2099-05-03T14:09:00	1.0	128.0	V L
MN.VLC..VHZ	2022-05-03T13:09:00	2099-05-03T14:09:00	1.0	128.0	V L
MN.VSL..VHE	2017-06-30T12:00:00	-	1.0	362.0	V L
MN.VSL..VHN	2017-06-30T12:00:00	-	1.0	362.0	V L
MN.VSL..VHZ	2017-06-30T12:00:00	-	1.0	362.0	V L
MN.WDD..VHE	2017-07-01T00:00:00	-	1.0	128.0	V L
MN.WDD..VHN	2017-07-01T00:00:00	-	1.0	128.0	V L
MN.WDD..VHZ	2017-07-01T00:00:00	-	1.0	128.0	V L
RO.CIOR..SHE	2013-06-18T00:00:00	-	100.0	1.0	S E
RO.CIOR..SHN	2013-06-18T00:00:00	-	100.0	1.0	S E
RO.CIOR..SHZ	2013-06-18T00:00:00	-	100.0	1.0	S E
SJ.SVIS..BHZ	2006-01-01T00:00:00	-	20.0	2.0	B S
SJ.SVIS..BHE	2006-01-01T00:00:00	-	20.0	2.0	B S
SJ.SVIS..BHN	2006-01-01T00:00:00	-	20.0	2.0	B S
CL.MALA.00.EHE	2022-12-03T12:30:00	2500-12-31T23:59:59	100.0	13.0	E H
CL.MALA.00.EHN	2022-12-03T12:30:00	2500-12-31T23:59:59	100.0	13.0	E H
CL.MALA.00.EHZ	2022-12-03T12:30:00	2500-12-31T23:59:59	100.0	13.0	E H
MT.CLP2.01.DH1	2021-05-14T14:00:00	2500-12-31T23:59:59	250.0	12.0	D C
MT.CLP2.01.DH3	2021-05-14T14:00:00	2500-12-31T23:59:59	250.0	12.0	D C
RO.BUC..SHE	2013-07-11T00:00:00	-	100.0	1.0	S E
RO.BUC..SHN	2013-07-11T00:00:00	-	100.0	1.0	S E
RO.BUC..SHZ	2013-07-11T00:00:00	-	100.0	1.0	S E
RO.SIBR..SHZ	2013-10-02T00:00:00	-	100.0	1.0	S E

Cross-checks of corner periods from metadata and independent list

[1]	[2]	[3]	[4]	[1]	[2]	[3]	[4]	[1]	[2]	[3]	[4]	[1]	[2]	[3]	[4]
BW.OBER	1	120	1	IV.CPTP	40	123	0	IV.CPIS	60	380	0	ME.CEME	120	31	0
BW.PART	1	120	1	IV.EFIU	40	123	0	IV.CPOZ	60	123	0	OX.MPRI	120	41	0
BW.RMOA	1	123	1	IV.ENIC	40	123	0	IV.CRTO	60	380	0	PL.NIE	120	97	0
BW.RNHA	1	123	1	IV.EPOZ	40	123	0	IV.CSFT	60	380	0	PL.OJC	120	97	0
BW.RNON	1	120	1	IV.ESML	40	123	0	IV.CSOB	60	123	0	RO.CHLR	120	102	0
BW.RTBE	1	120	1	IV.GALF	40	123	0	IV.IFOR	60	380	0	RO.NEGR	120	102	0
BW.RTSH	1	120	1	IV.HMDC	40	758	0	IV.IMTC	60	380	0	RO.PETR	120	102	0
BW.UH3	1	120	1	IV.HVZN	40	123	0	IV.IOCA	60	117	0	RO.SLCR	120	102	0
BW.ZUGS	1	123	1	IV.IST3	40	758	0	IV.VBKN	60	380	0	RO.BAIL	120	102	0
CH.FLACH	1	123	1	IV.MTSN	40	117	0	IV.VCRE	60	380	0	RO.BANR	120	102	0
IV.BOZZ	1	123	1	IV.OPPE	40	123	0	IV.VTIR	60	380	0	RO.BOSR	120	102	0
AC.PUK	30	41	0	IV.PETRA	40	123	0	IV.VVDG	60	380	0	RO.CICN	120	102	0
BS.JMB	30	3	1	IV.RAFF	40	123	0	SL.VOJS	60	123	0	RO.EFOR	120	102	0
HA.ACOR	30	123	0	IV.RESU	40	123	0	GU.RORO	100	47	0	RO.HUMR	120	102	0
HA.AXAR	30	123	0	IV.SSY	40	123	0	HT.KNT	100	208	0	RO.LEHL	120	102	0
HA.MDRA	30	63	0	OX.ACOM	40	125	0	IV.MMGO	100	123	0	RO.MANR	120	102	0
HA.VILL	30	123	0	RO.RASA	50	102	0	NI.VINO	100	41	0	RO.NEHR	120	102	0
HL.LKR	30	123	0	RO.COPA	50	102	0	RO.BIZ	119	40	0	RO.NGRR	120	102	0
IV.CAVT	30	123	0	RO.DRGR	50	102	0	RO.ODBI	119	40	0	RO.PANC	120	102	0
IV.CRJA	30	123	0	RO.IASR	50	102	0	RO.TPGR	119	40	0	RO.SAHR	120	102	0
IV.IPSM	30	190	0	RO.PUNG	50	102	0	AC.SRN	120	41	0	RO.SRE	120	102	0
IV.IVLC	30	190	0	RO.SULR	50	102	0	BS.BLKB	120	102	0	RO.TLCR	120	102	0
IV.MFNL	30	758	0	RO.VLAD	50	102	0	BS.ELND	120	102	0	RO.ZIMR	120	102	0
IV.MTGR	30	41	0	RO.ARR	59	30	0	BS.KALB	120	102	0	SK.LANS	120	1	1
IV.PTMR	30	190	0	RO.BZS	59	30	0	BS.KUBB	120	102	0	SK.ZST	120	20	0
IV.VISG	30	190	0	CR.BRJN	60	30	0	BS.LOZB	120	102	0	SL.BOJS	120	380	0
IV.VMGN	30	190	0	CR.HVAR	60	123	0	BS.PLVB	120	102	0	IV.FIR	240	120	0
KO.PHSR	30	123	0	CR.KALN	60	30	0	BS.RAZG	120	102	0	IV.MSSA	240	47	0
NI.POLC	30	190	0	CR.MOSL	60	30	0	BS.SHAB	120	102	0	RO.BURAR	333	645	0
RF.GEPF	30	190	0	CR.PLIT	60	30	0	CZ.KRLC	120	60	0	SL.GORS	360	123	0
RO.LEOM	30	1	1	GU.GRAM	60	380	0	CZ.OKC	120	30	0	AC.PHP	120	41	0
RO.PLOR	30	1	1	GU.NEGI	60	380	0	HU.TIH	120	61	0	AC.LSK	120	380	0
SK.VYHS	30	123	0	GU.RNCA	60	380	0	IV.EMAS	120	758	0				
AC.BCI	40	380	0	HC.KNDR	60	123	0	IV.FIAM	120	47	0				
HP.AXS	40	123	0	HL.NVR	60	123	0	IV.HCRL	120	758	0				
HP.DSF	40	117	0	HL.RDO	60	123	0	IV.HSRS	120	758	0				
IV.CMDO	40	123	0	IV.CAAM	60	123	0	IV.HVIT	120	758	0				
IV.CORL	40	123	0	IV.CFMN	60	123	0	KO.ERIK	120	30	0				

[1] Station

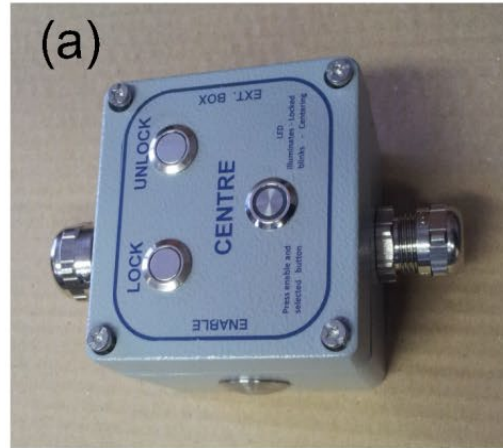
[2] Corner period [s] – from Petr's table

[3] Corner period [s] – from metadata

[4] Change BB<->SP?

Developed control and calibration units (IG CAS)

CMG host box



CMG control and calibration unit



CMG centring unit



STS2 control and calibration unit



GAIA gain and calibration unit

